Continuous coupling trajectories — run report

RUN

215311_cross-sweep_phi1-rerun
Status: completed | Started: 2026-09-29T20:53:11.509017+00:00

DIAGNOSTIC SUMMARY

Target actuator(s): phi1; sine: 0.1164 Hz; measured PAX rate: 6.87 Hz; saved RP reference rate: 381 Hz; estimated PAX samples/cycle: 59.00.
Recorded cycles: 3; bias values: phi1 target 11, phi2 target 0. Waveform tracking: correct within descriptive threshold. Rising/falling branches (>5° descriptive threshold): strong separation present. Cycle repeatability (5° descriptive threshold): nonrepeatable above threshold. Bias-dependent plane changes (>5° descriptive threshold): obvious changes present. Low-DoP intervals: True. These thresholds are flags, not mechanism assignments.

PURPOSE

Continuously sine-drive phi1, phi2, or both while stepping and settling the orthogonal bias; record PAX plus measured OUT1/OUT2 references.

REQUIRED OPTICAL SETUP

PAX at final output; both paths open unless prompted. Wire OUT1 to IN1 and OUT2 to IN2 as voltage references. 'both' runs the complete diagnostic for both actuators automatically.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port
Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

RUN COMMENT

another re run for the sake of having data to analyze later. time syncing will be done during analysis for now until further debugged. RP LV is set on both IN1 and IN2 therefore reference command voltage should be faithful.

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

Continuous coupling trajectories — run report

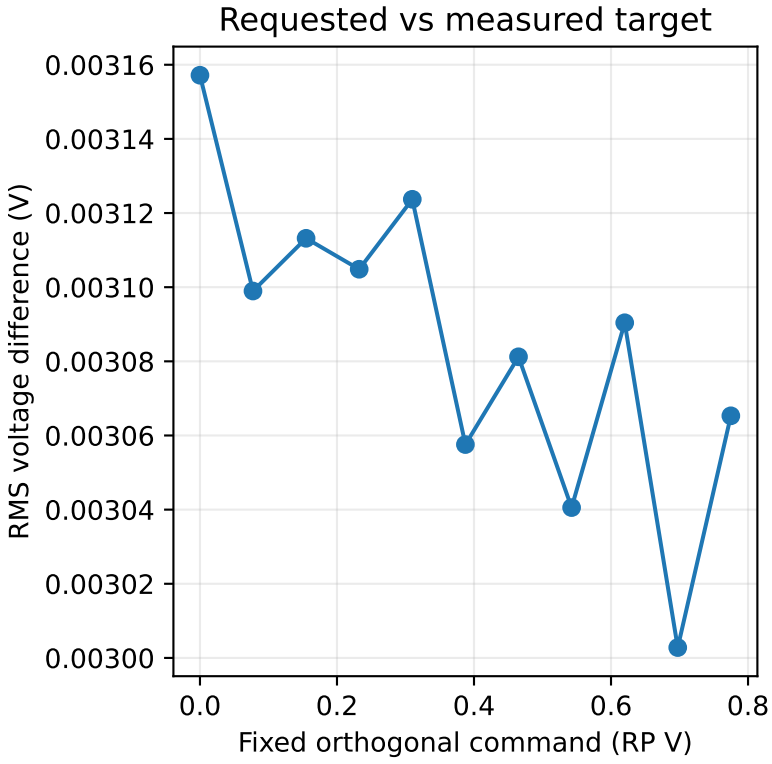
DATA

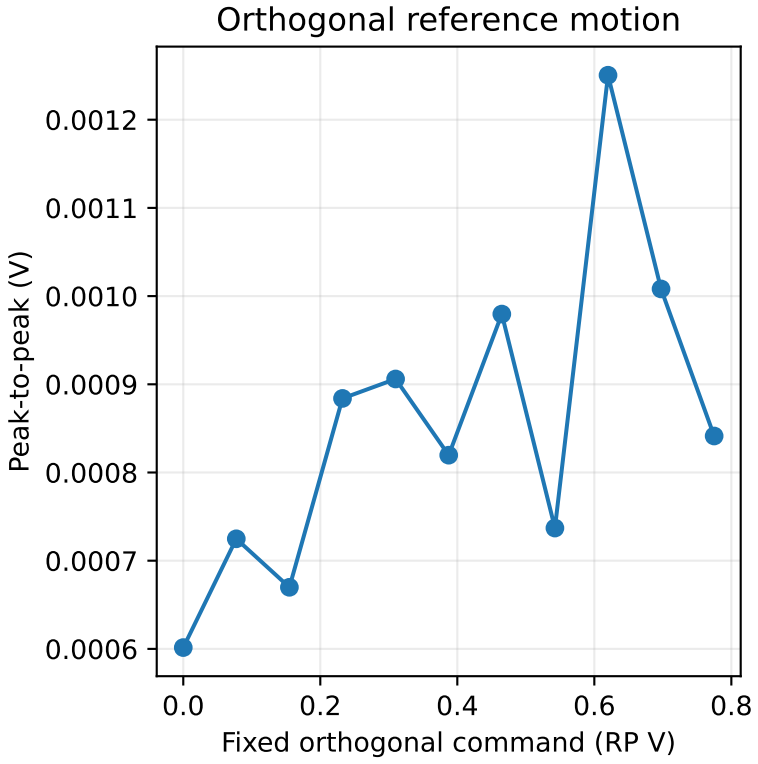
2,680 recorded samples; acquisition: 2,680
Source: data.csv; full acquisition settings: recipe.json; status: run.json

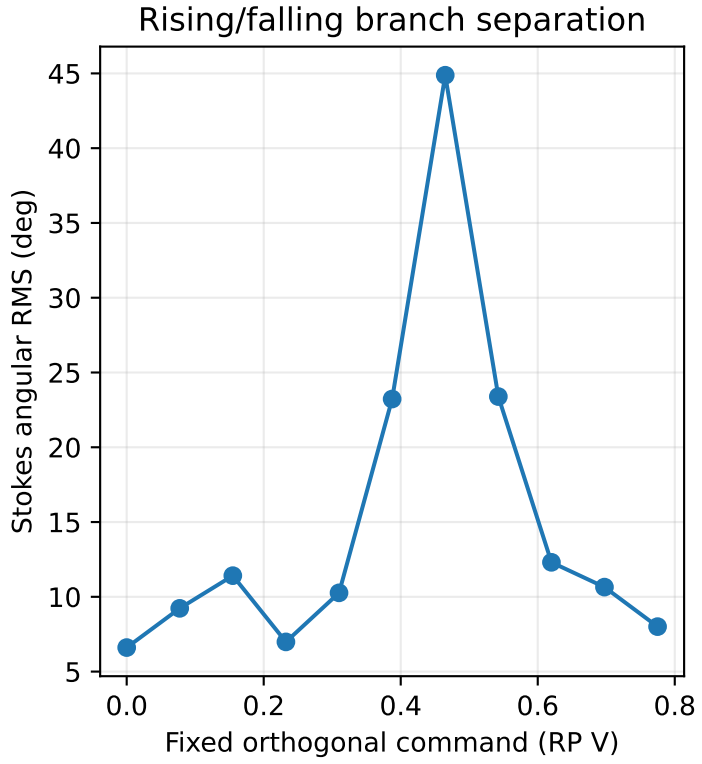
RAW DOP

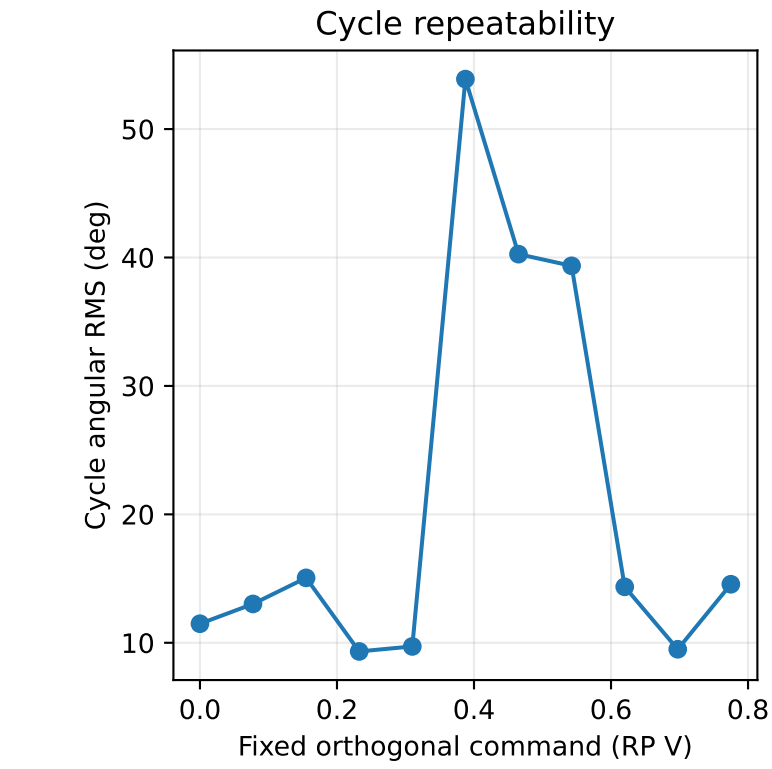
Mean 0.9203; range 0.2613 to 1.0454. 5 samples outside 0–1. Telemetry is not a new calibration.

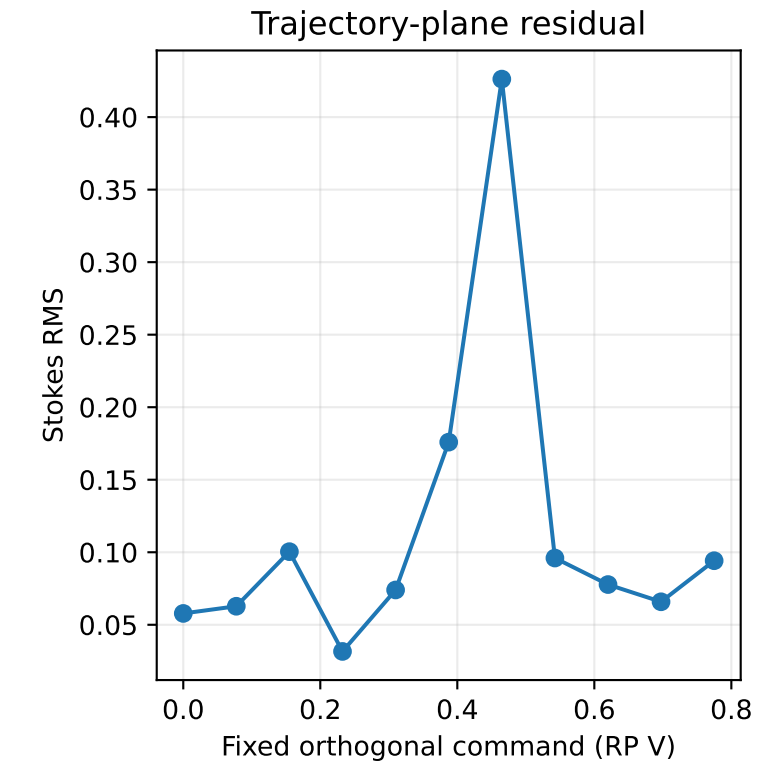
Continuous phi1 diagnostic — bias dependence

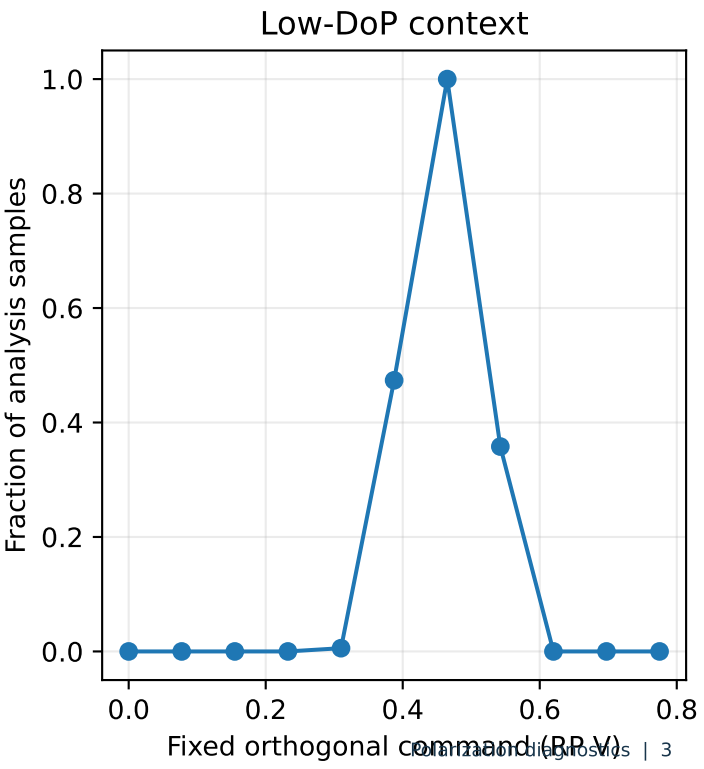




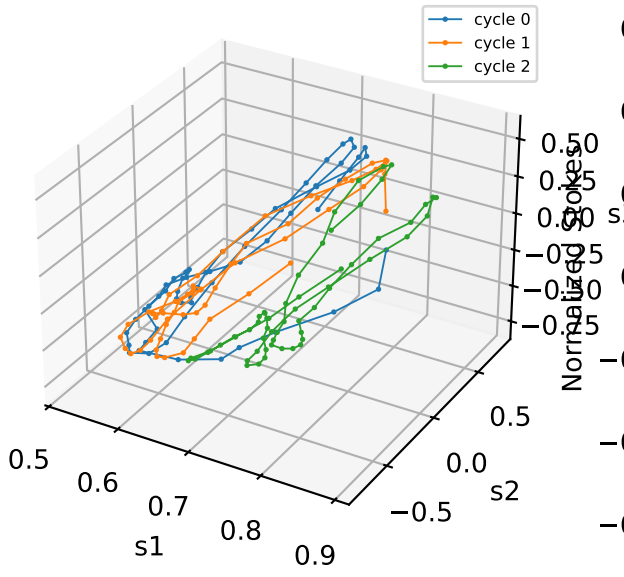


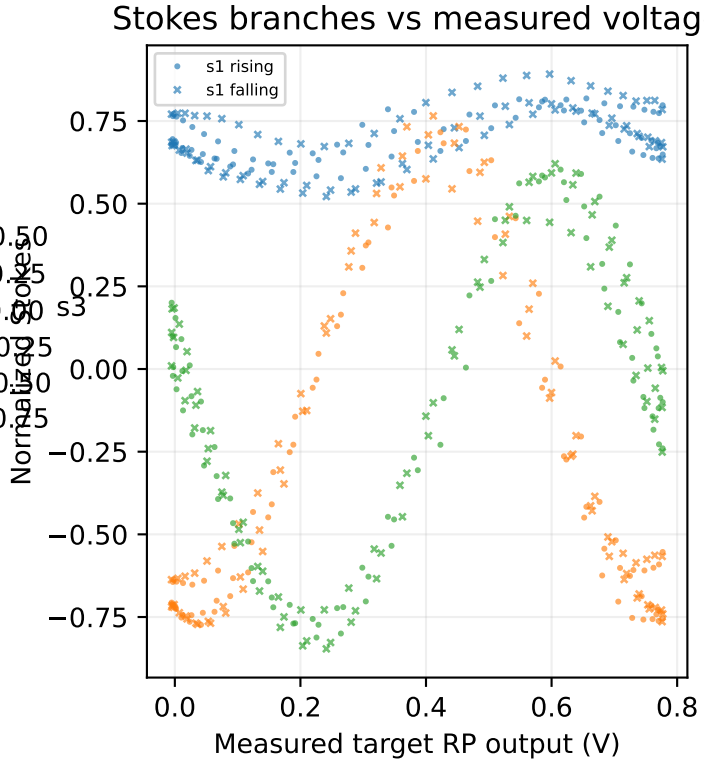


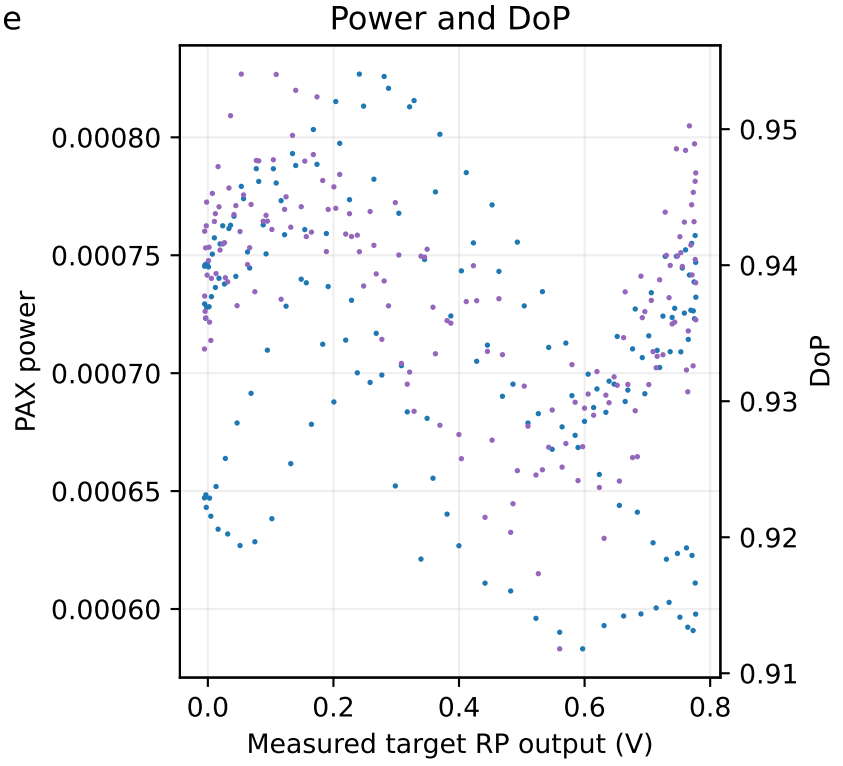


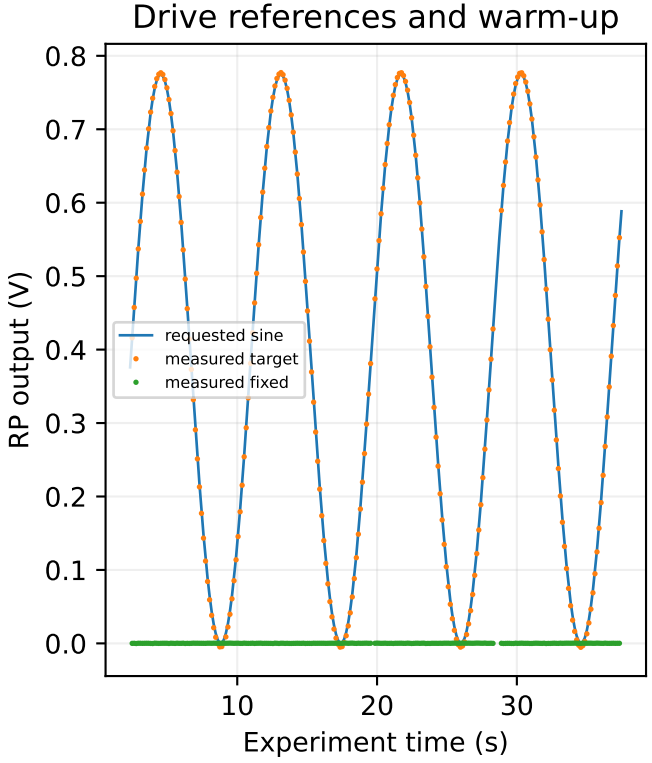


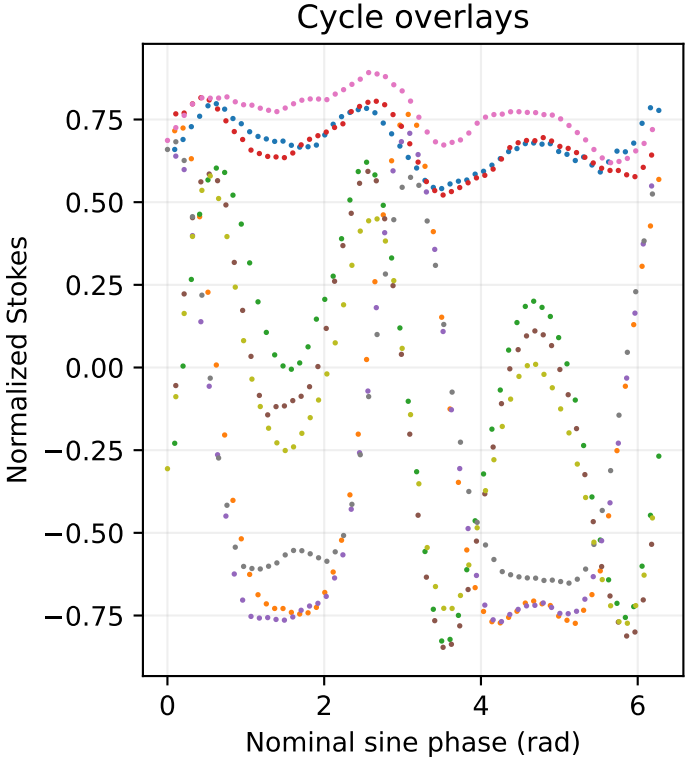
phi1 sine; fixed phi2 = 0 RP V (segment 0)





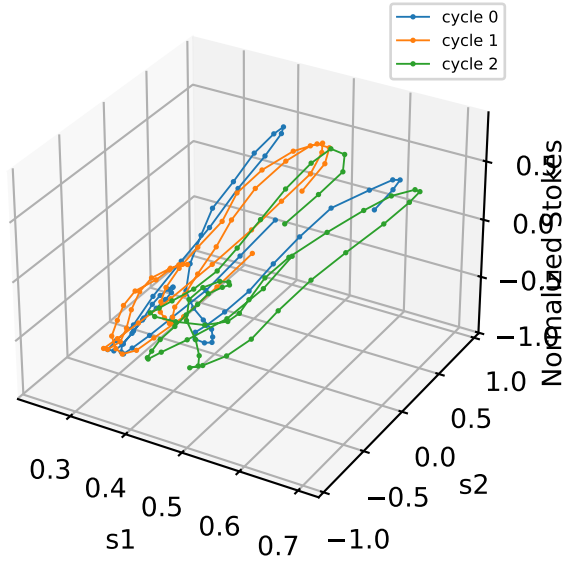


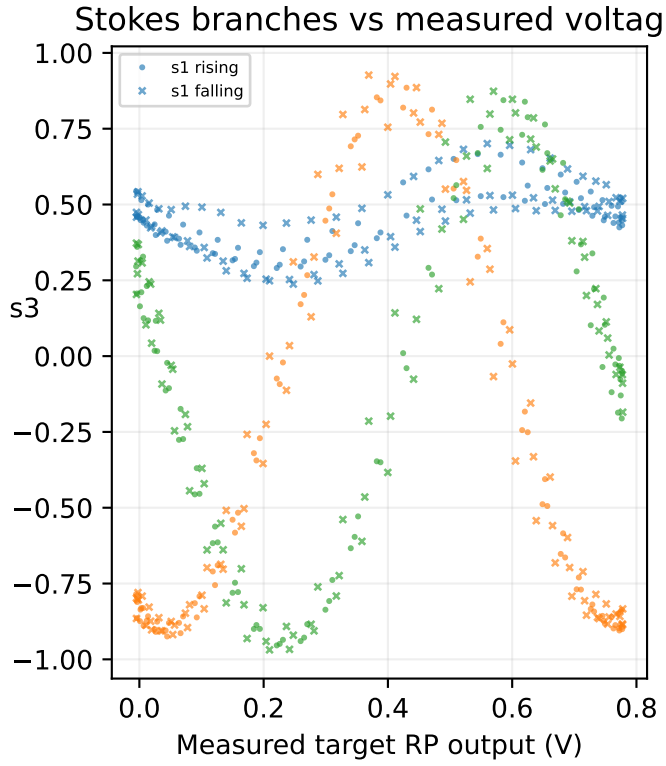


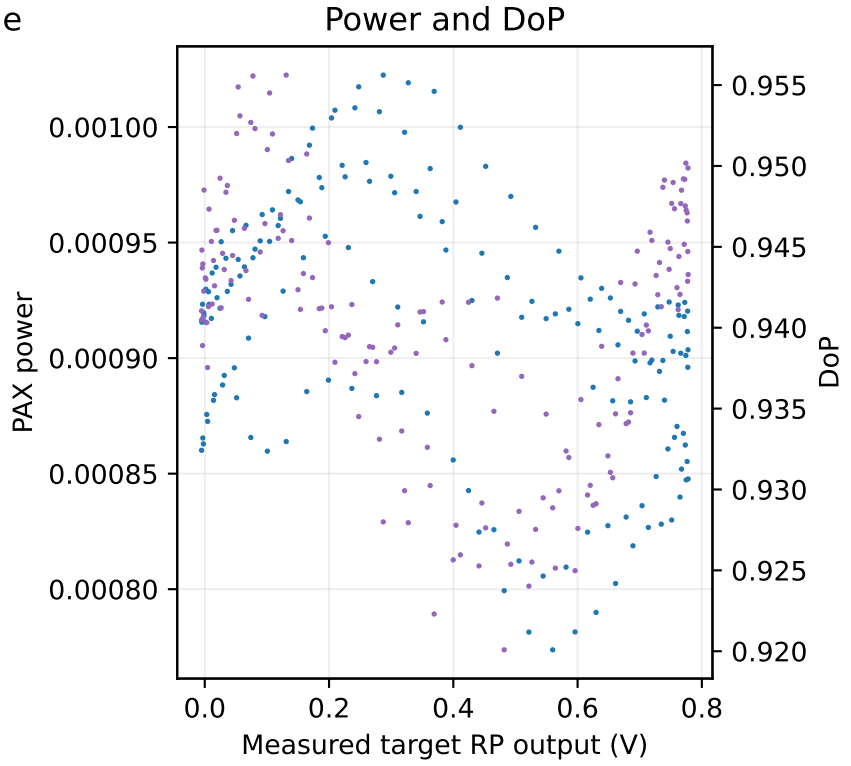


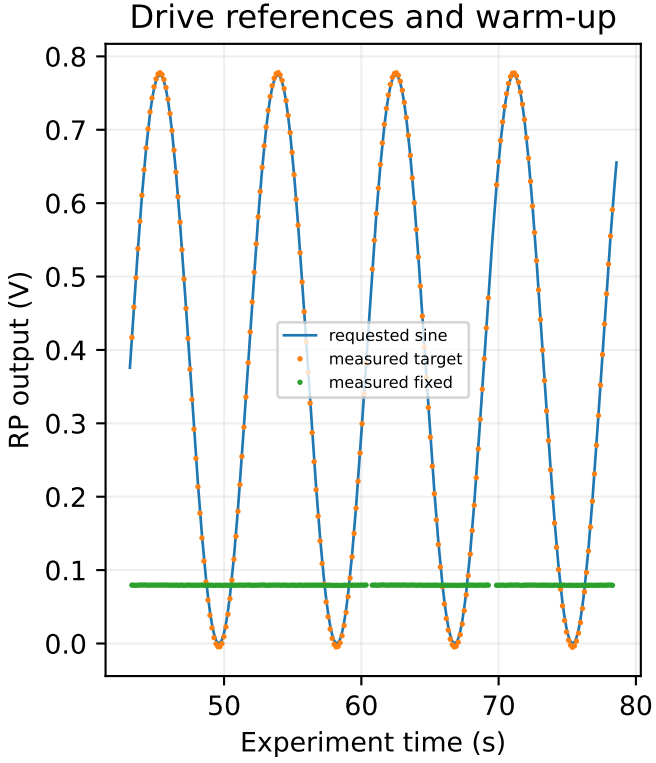
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.989, -0.026, -0.144]
Plane residual RMS: 0.05782
Rising/falling angular RMS: 6.61°
Rising/falling power RMS: 6.03e-05 PAX units
Cycle angular RMS: 11.5°
Cycle power RMS: 7.92e-05 PAX units
Cycle DoP RMS: 0.00513
Tracking RMS: 0.003157 V
Reference coverage: 97.7%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, -0.03, -0.13], residual 0.0411
Cycle 1 plane: [0.98, -0.04, -0.17], residual 0.0191
Cycle 2 plane: [0.99, 0.01, -0.16], residual 0.0317

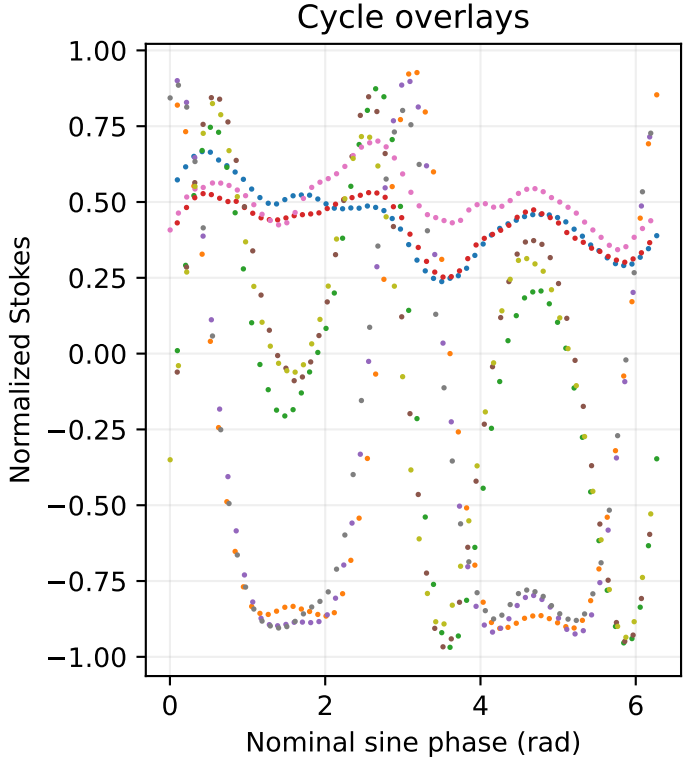
phi1 sine; fixed phi2 = 0.07739 RP V (segment 1)





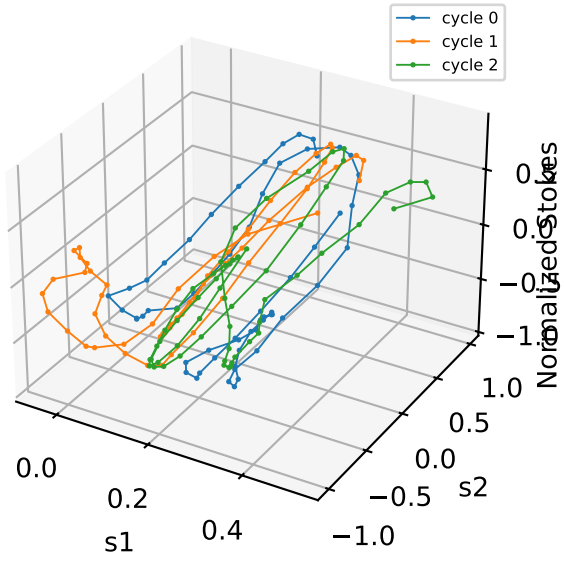


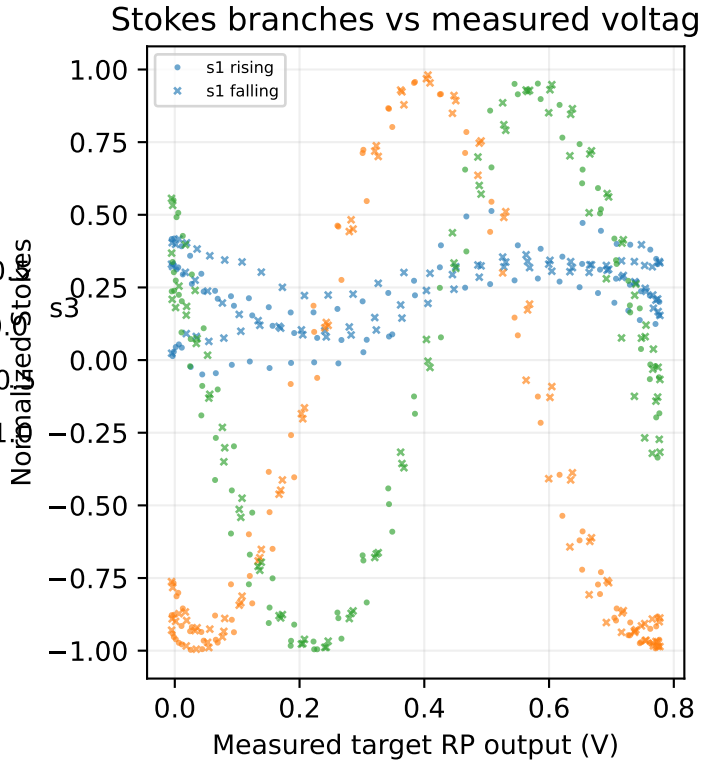


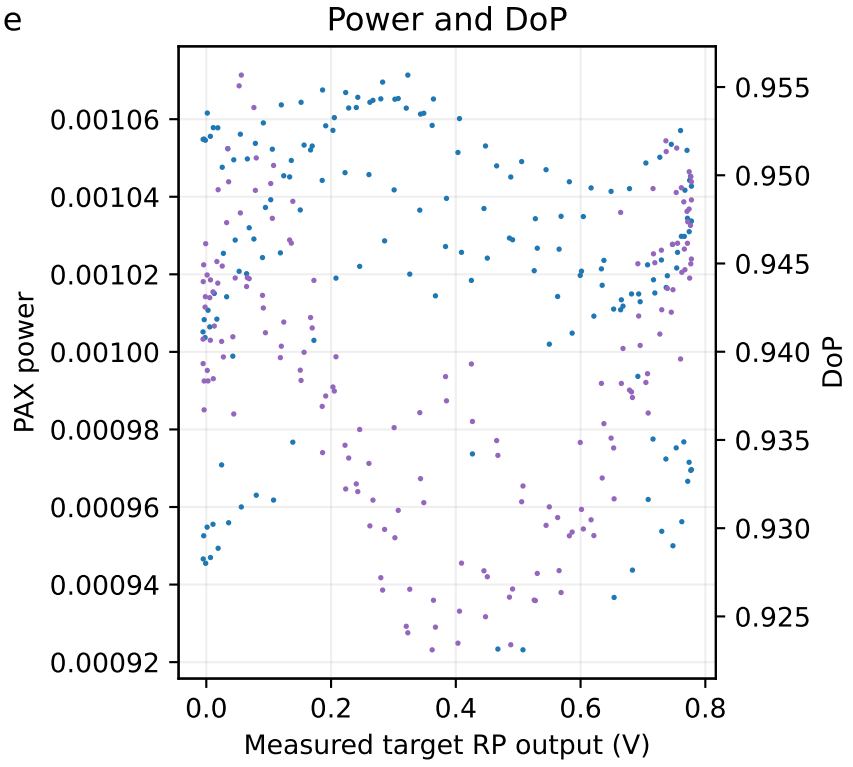


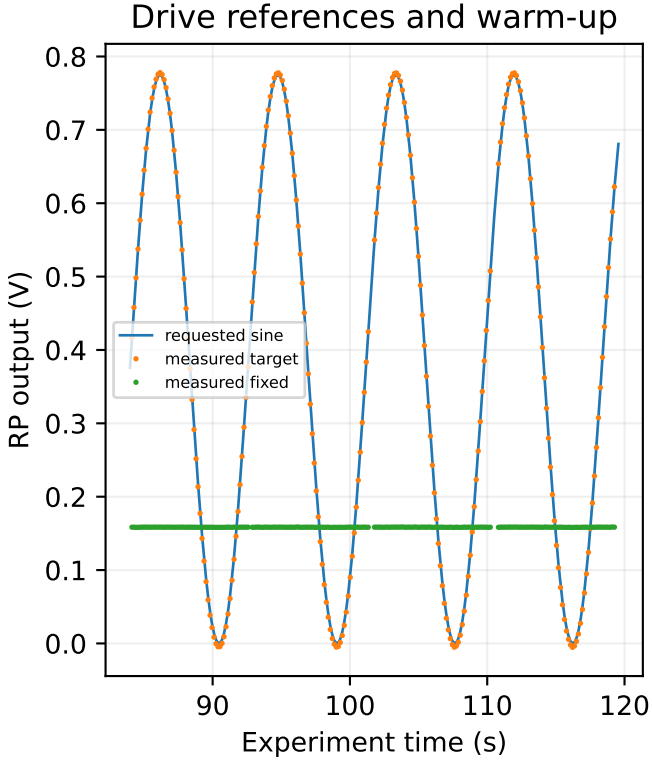
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.989, -0.005, -0.147]
Plane residual RMS: 0.06277
Rising/falling angular RMS: 9.23°
Rising/falling power RMS: 6.62e-05 PAX units
Cycle angular RMS: 13°
Cycle power RMS: 7.1e-05 PAX units
Cycle DoP RMS: 0.0038
Tracking RMS: 0.003099 V
Reference coverage: 97.2%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, 0.02, -0.17], residual 0.0651
Cycle 1 plane: [0.99, -0.00, -0.14], residual 0.022
Cycle 2 plane: [0.99, -0.03, -0.14], residual 0.0469

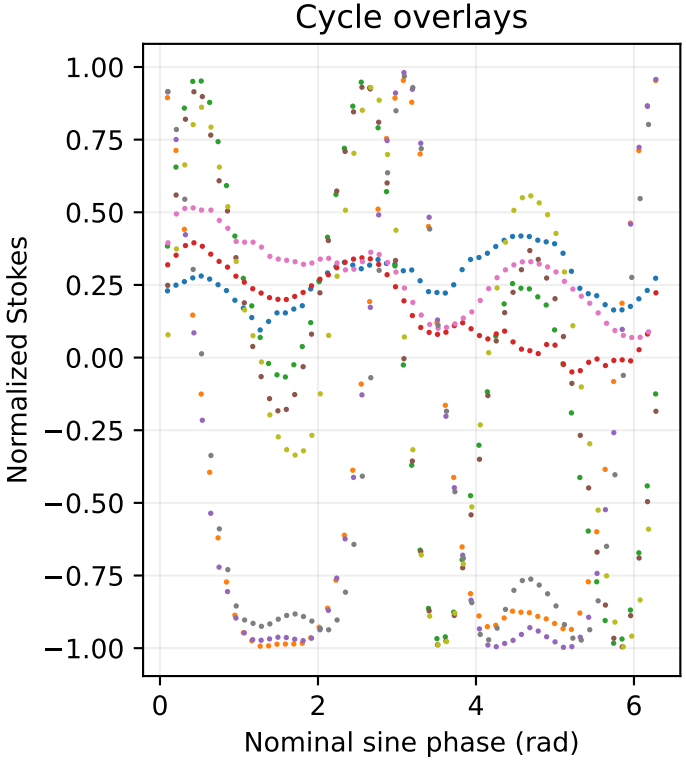
phi1 sine; fixed phi2 = 0.1549 RP V (segment 2)





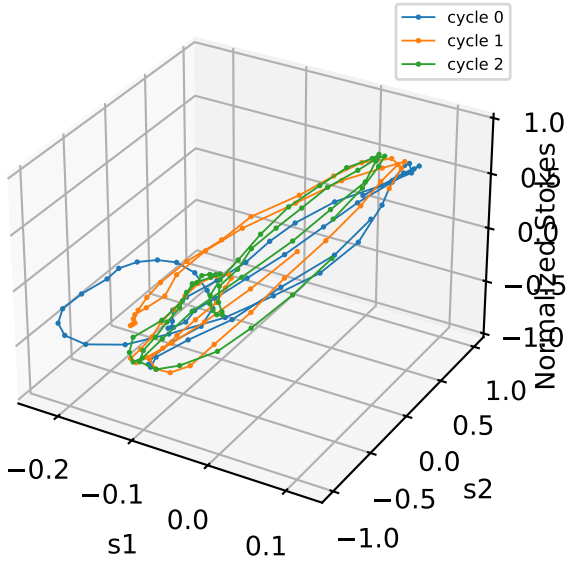


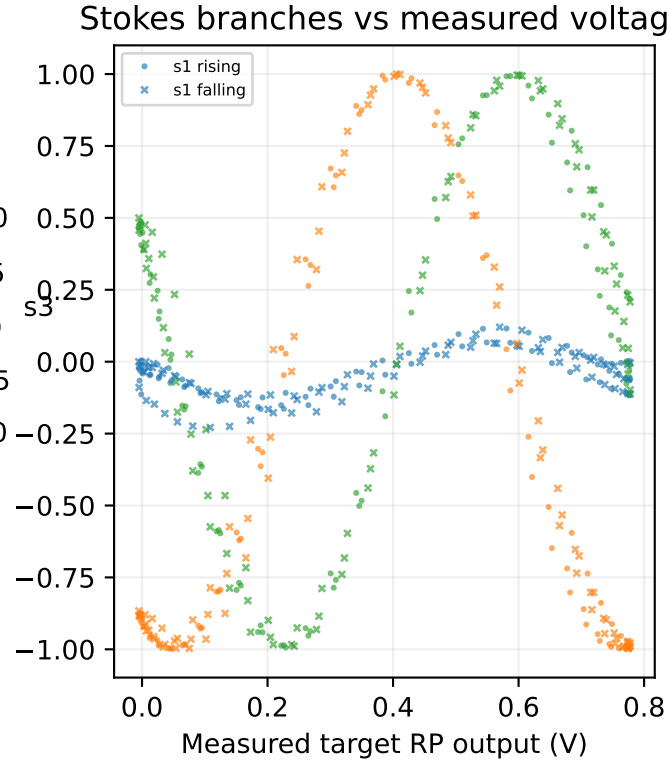


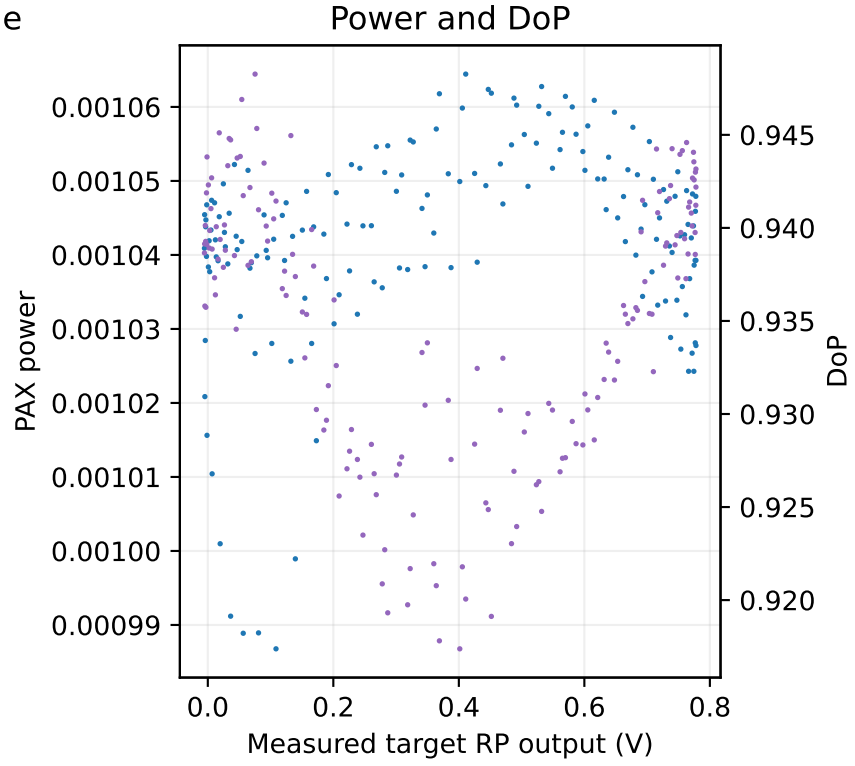


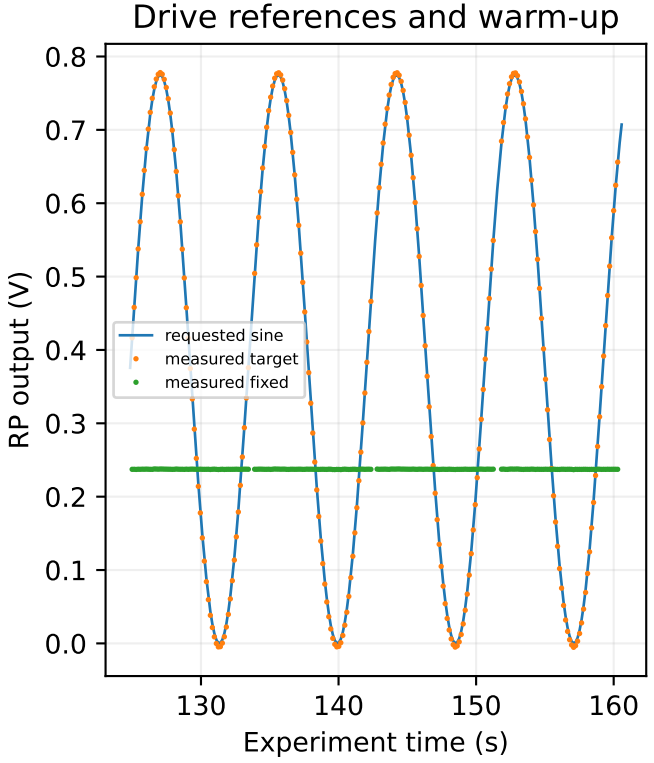
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.991, -0.014, -0.135]
Plane residual RMS: 0.1004
Rising/falling angular RMS: 11.4°
Rising/falling power RMS: 3.7e-05 PAX units
Cycle angular RMS: 15.1°
Cycle power RMS: 5.84e-05 PAX units
Cycle DoP RMS: 0.00322
Tracking RMS: 0.003113 V
Reference coverage: 96.6%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [1.00, 0.00, -0.05], residual 0.0767
Cycle 1 plane: [0.98, -0.05, -0.18], residual 0.0836
Cycle 2 plane: [0.98, 0.01, -0.18], residual 0.0698

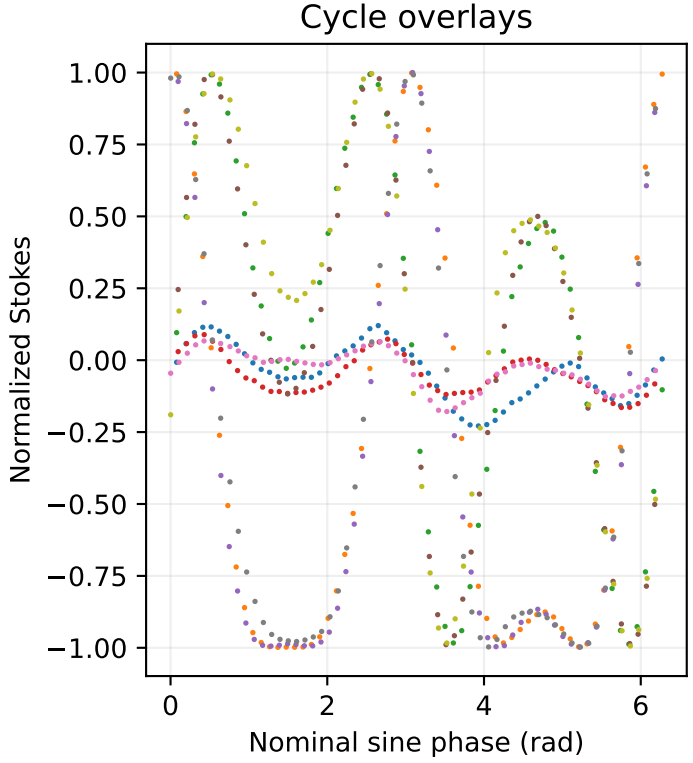
phi1 sine; fixed phi2 = 0.2324 RP V (segment 3)





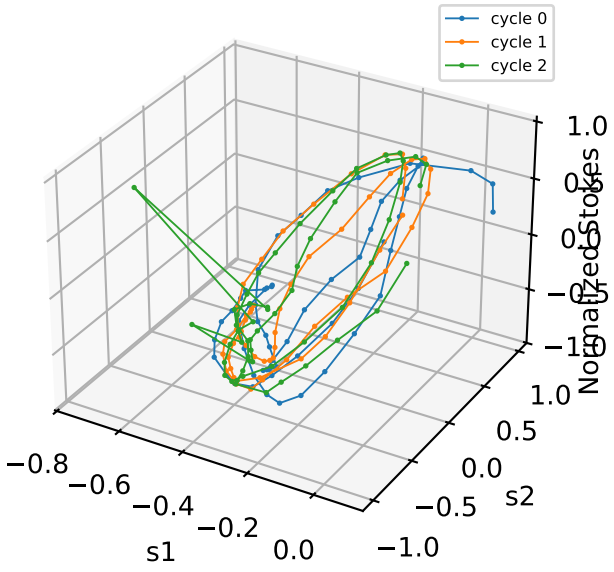


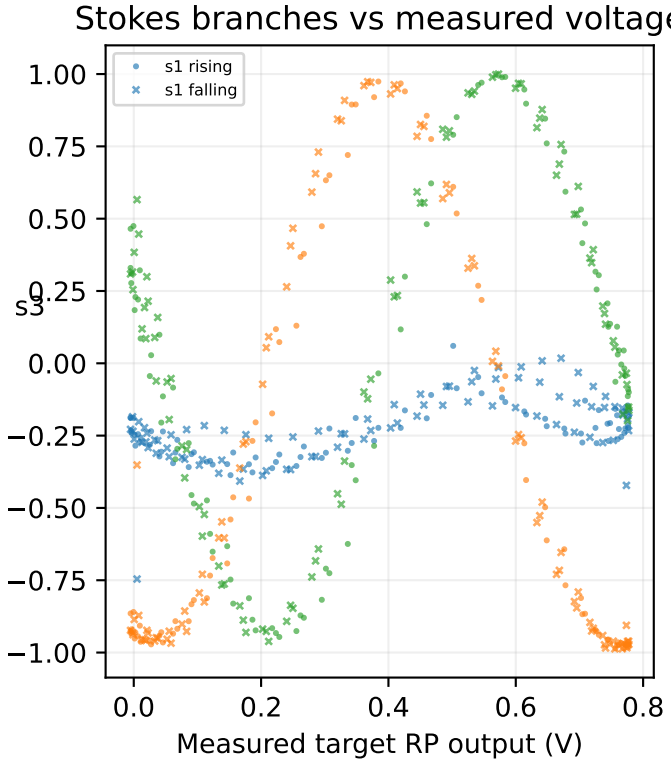


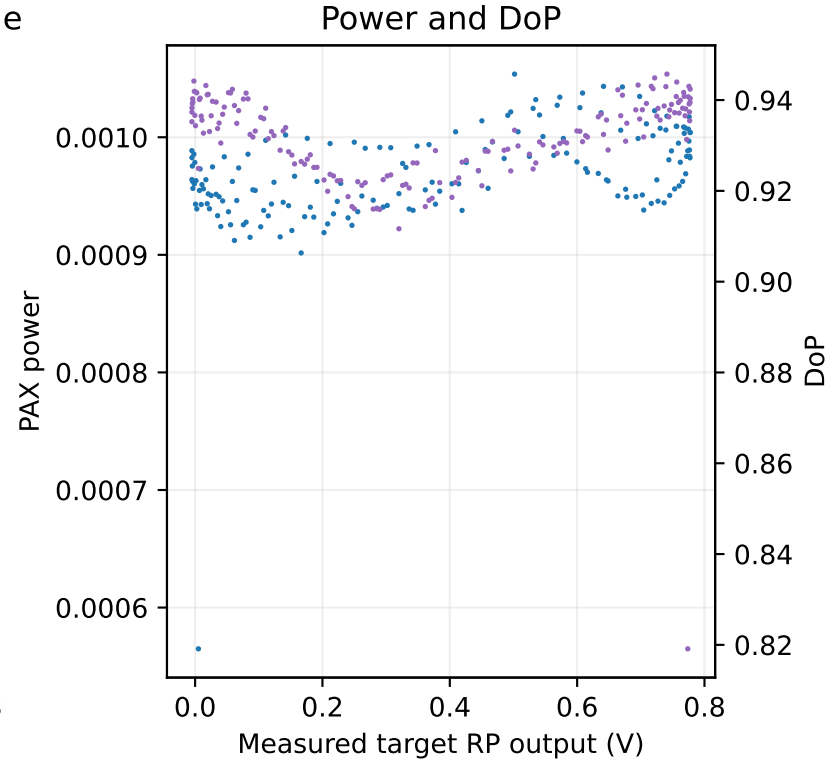


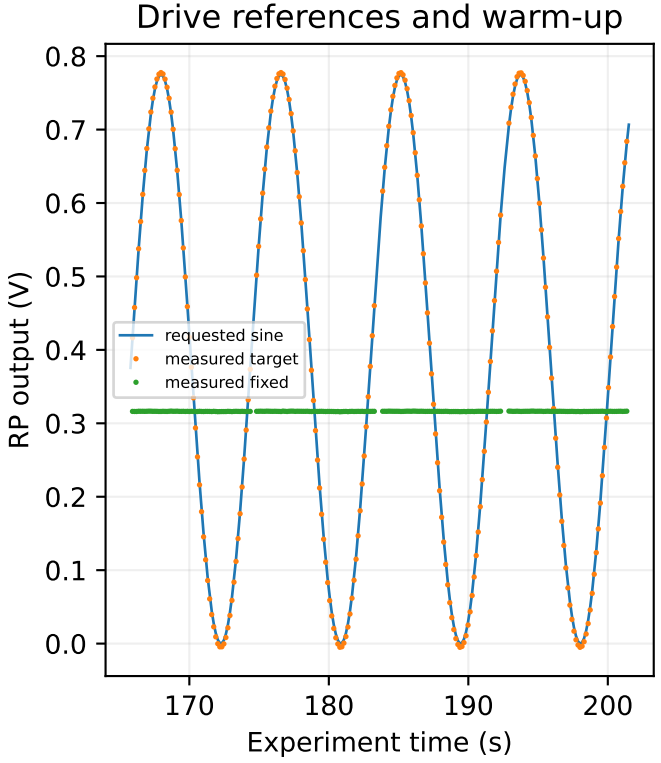
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.993, -0.038, -0.114]
Plane residual RMS: 0.03162
Rising/falling angular RMS: 6.99°
Rising/falling power RMS: 1.68e-05 PAX units
Cycle angular RMS: 9.33°
Cycle power RMS: 1.79e-05 PAX units
Cycle DoP RMS: 0.00266
Tracking RMS: 0.003105 V
Reference coverage: 96.0%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, -0.06, -0.13], residual 0.0403
Cycle 1 plane: [0.99, -0.04, -0.10], residual 0.0211
Cycle 2 plane: [0.99, -0.02, -0.10], residual 0.0193

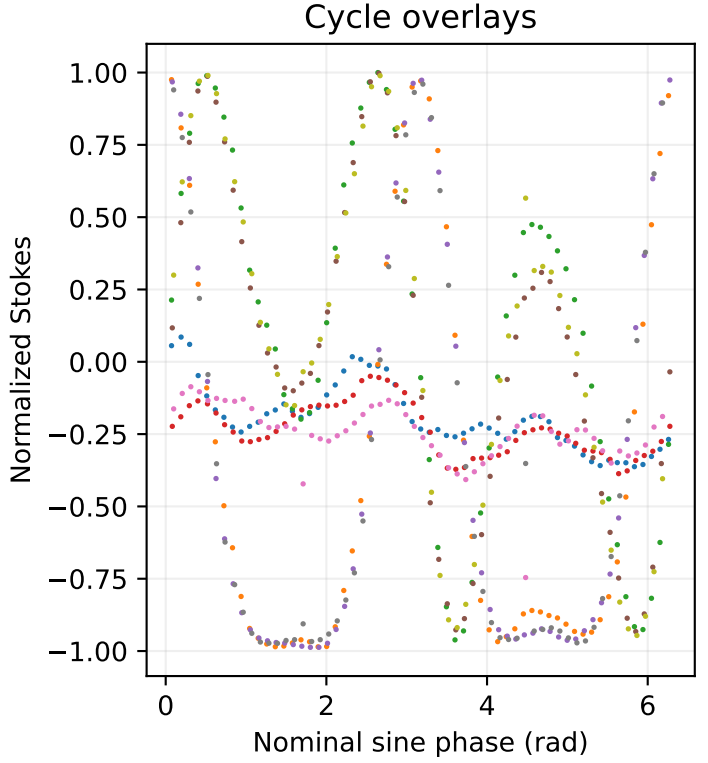
phi1 sine; fixed phi2 = 0.3099 RP V (segment 4)





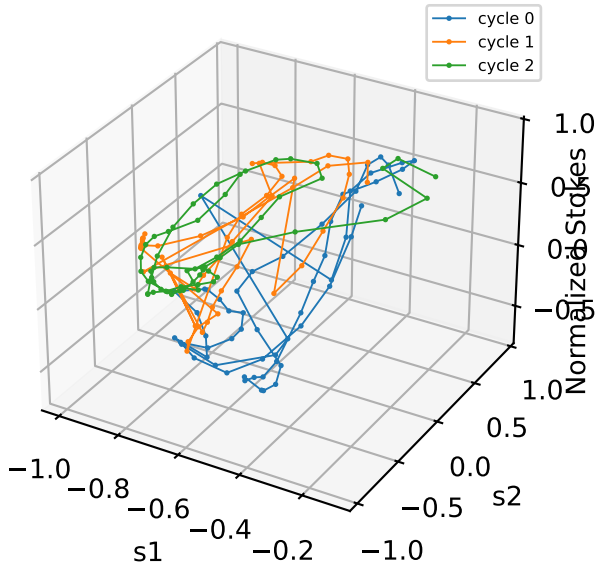


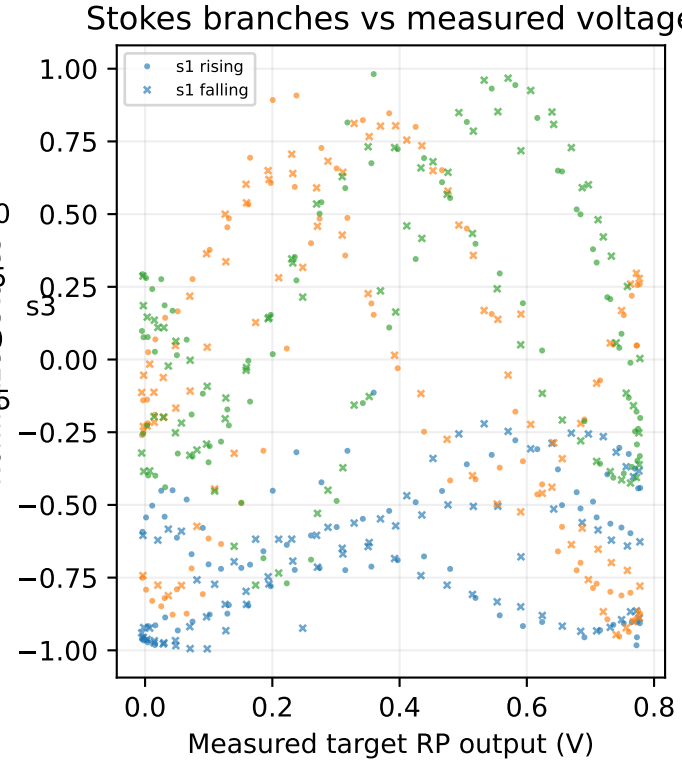


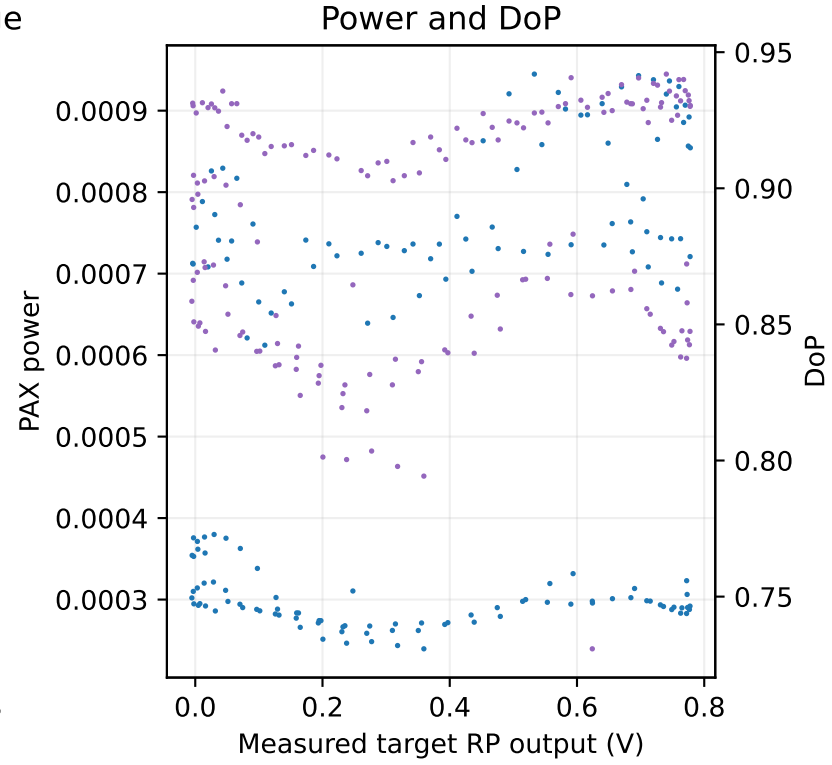


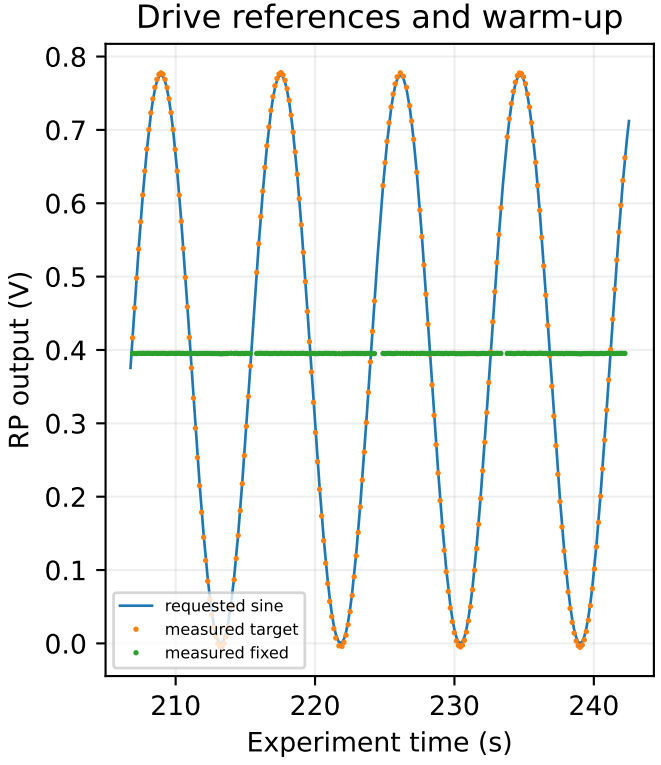
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.992, -0.030, -0.126]
Plane residual RMS: 0.074
Rising/falling angular RMS: 10.3°
Rising/falling power RMS: 4.8e-05 PAX units
Cycle angular RMS: 9.72°
Cycle power RMS: 5.29e-05 PAX units
Cycle DoP RMS: 0.00991
Tracking RMS: 0.003124 V
Reference coverage: 95.5%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, -0.04, -0.14], residual 0.076
Cycle 1 plane: [0.99, -0.02, -0.14], residual 0.0437
Cycle 2 plane: [0.99, -0.03, -0.10], residual 0.0839

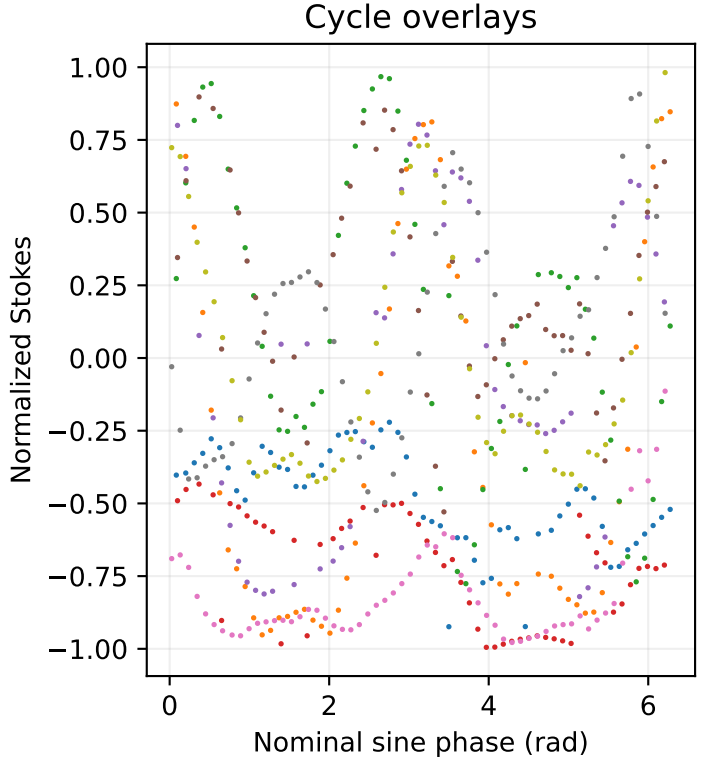
phi1 sine; fixed phi2 = 0.3875 RP V (segment 5)





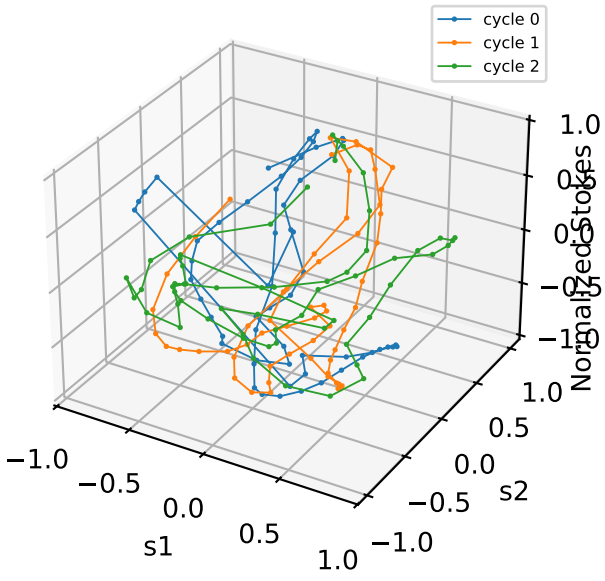


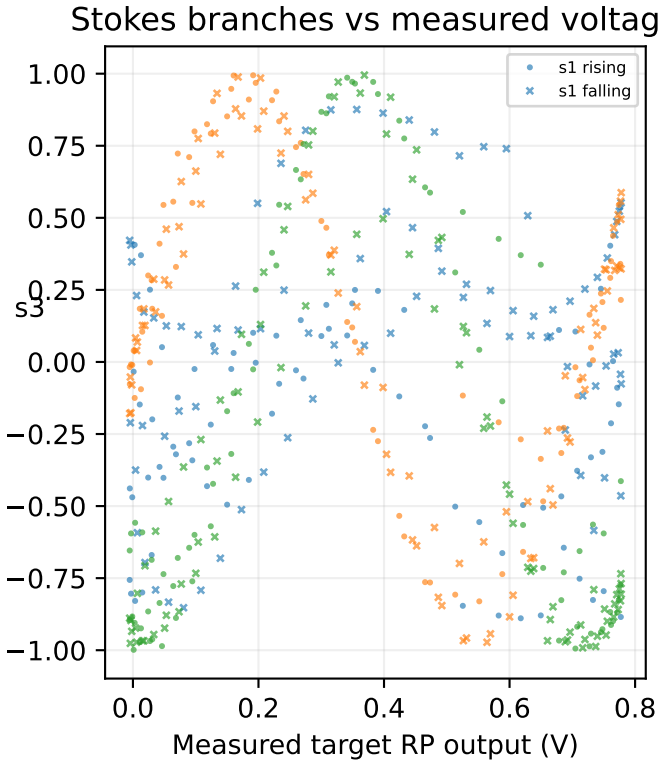


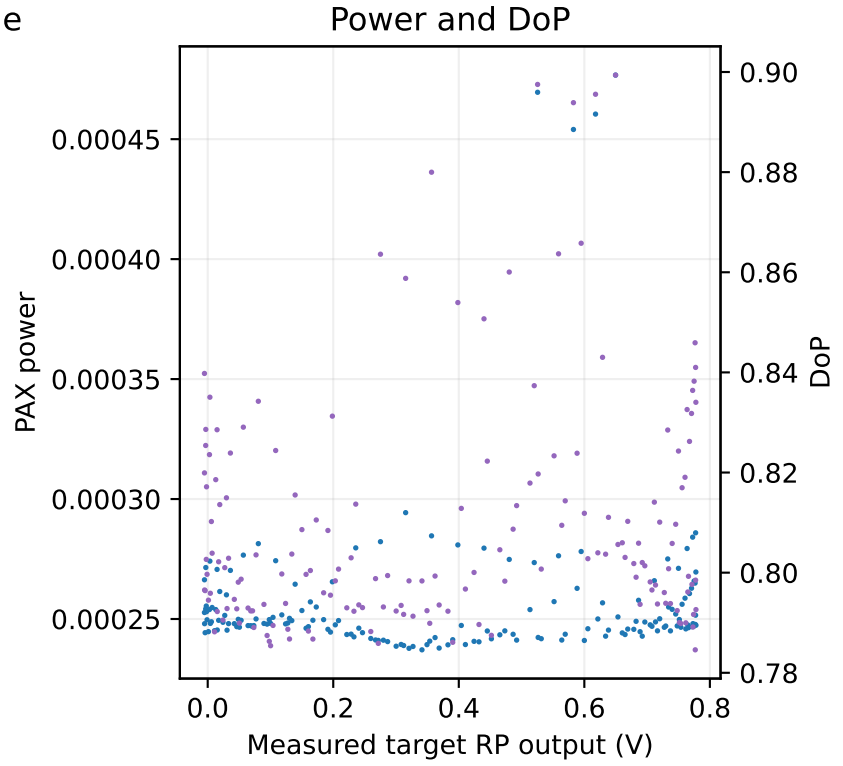


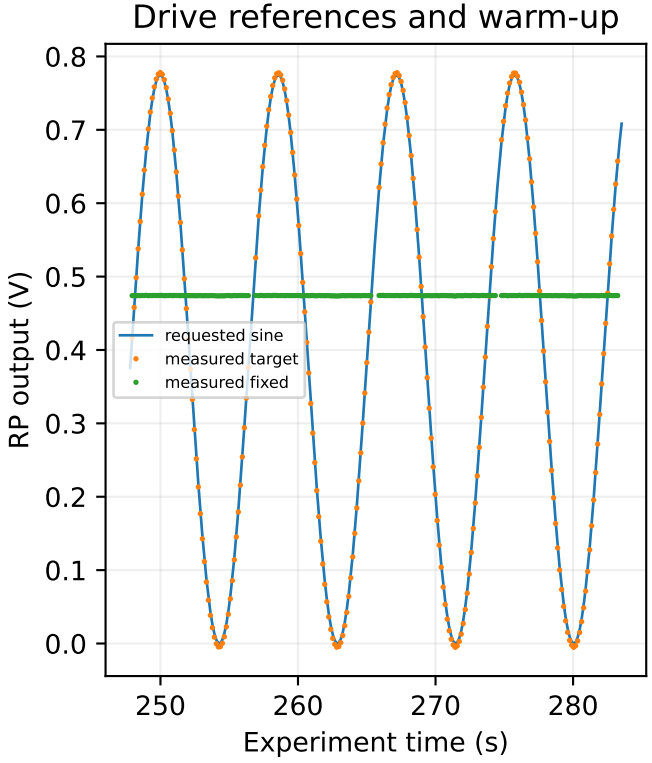
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.945, 0.061, -0.320]
Plane residual RMS: 0.1759
Rising/falling angular RMS: 23.2°
Rising/falling power RMS: 0.000146 PAX units
Cycle angular RMS: 53.9°
Cycle power RMS: 0.000415 PAX units
Cycle DoP RMS: 0.0642
Tracking RMS: 0.003058 V
Reference coverage: 96.5%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.97, 0.04, -0.23], residual 0.116
Cycle 1 plane: [0.94, 0.03, -0.35], residual 0.139
Cycle 2 plane: [0.92, -0.24, -0.31], residual 0.0753

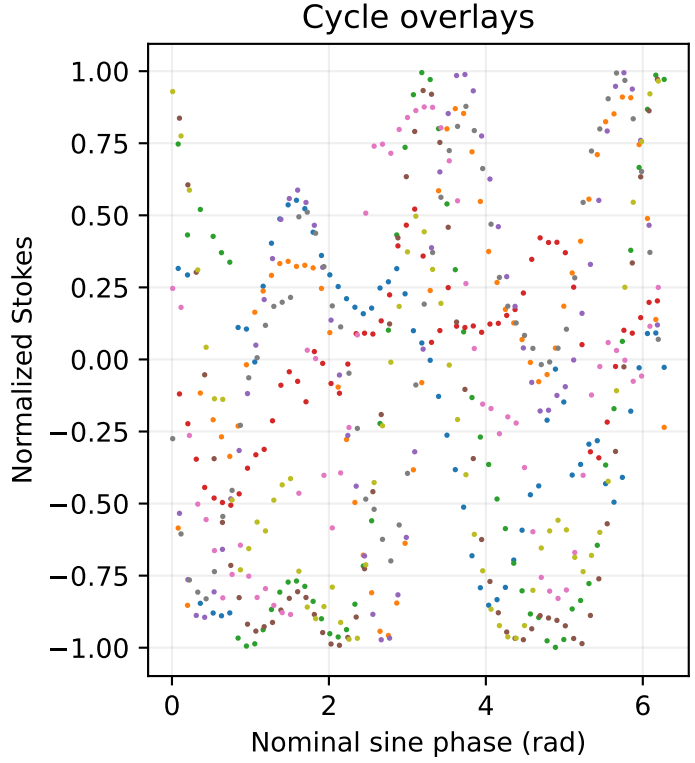
phi1 sine; fixed phi2 = 0.465 RP V (segment 6)





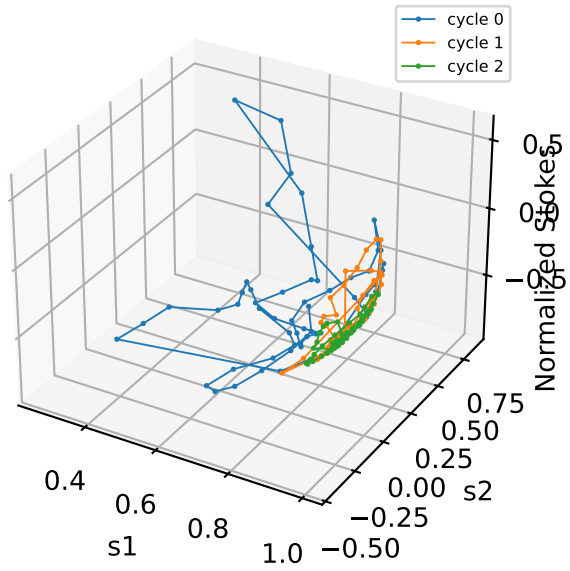


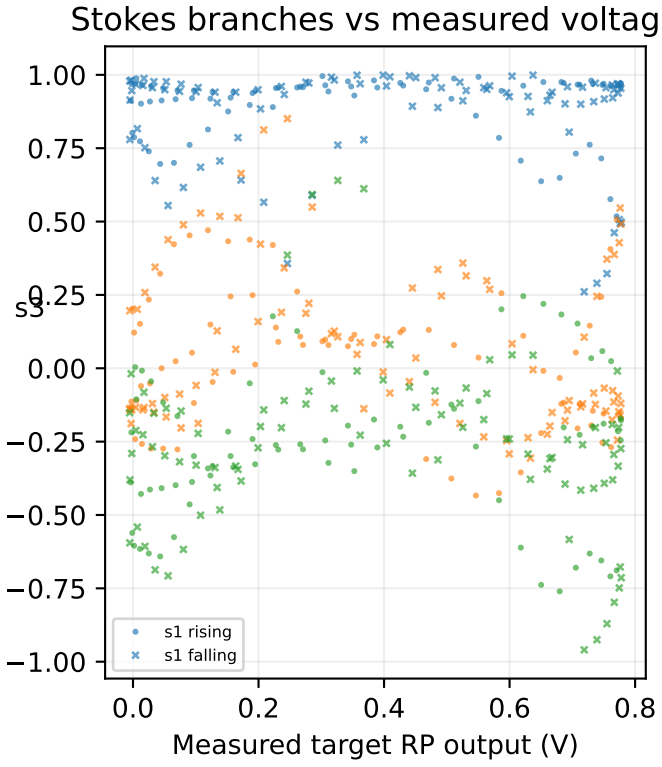


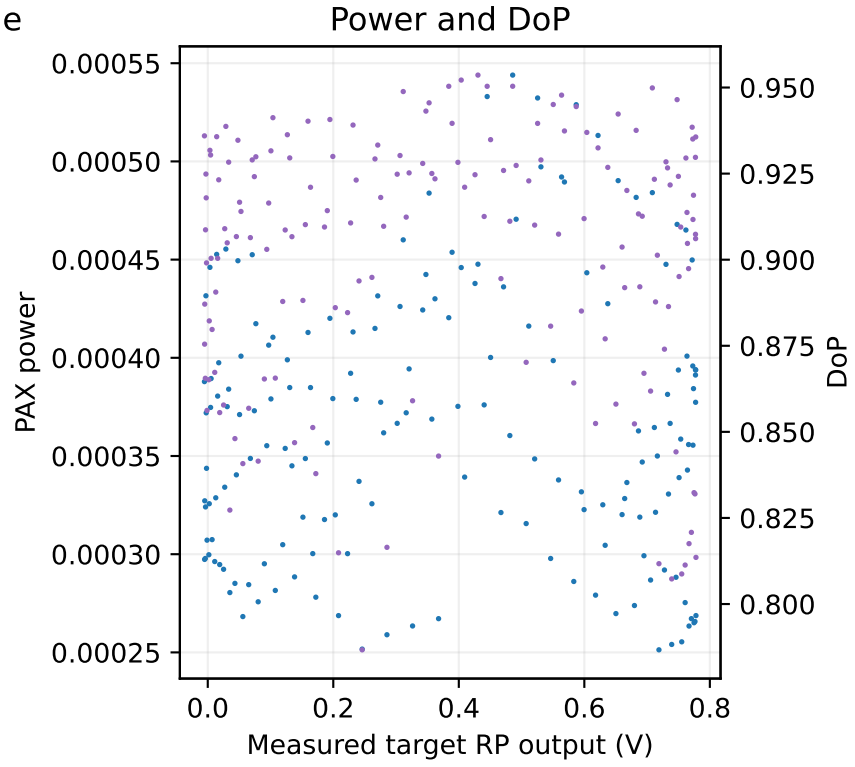


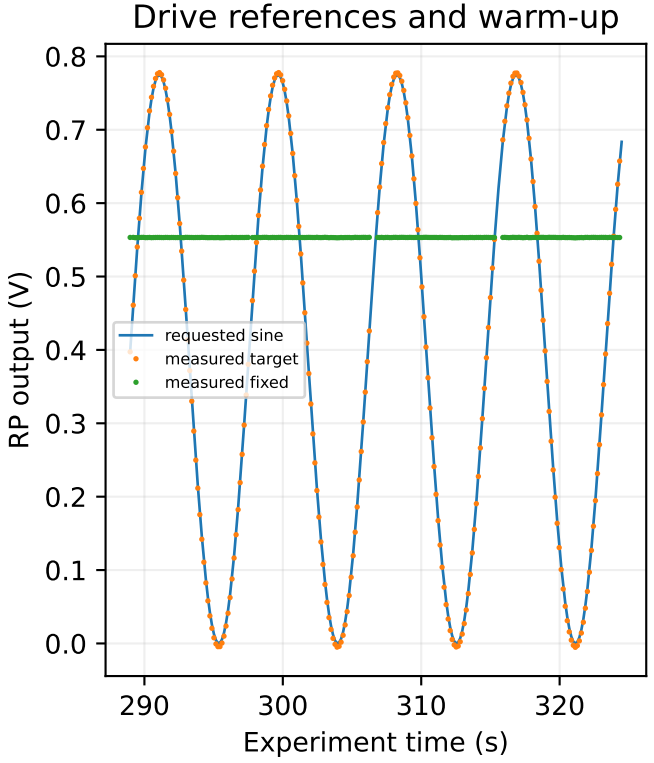
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.980, 0.063, -0.190]
Plane residual RMS: 0.4262
Rising/falling angular RMS: 44.9°
Rising/falling power RMS: 6.36e-05 PAX units
Cycle angular RMS: 40.3°
Cycle power RMS: 5.83e-05 PAX units
Cycle DoP RMS: 0.0333
Tracking RMS: 0.003081 V
Reference coverage: 96.0%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.85, 0.51, 0.12], residual 0.373
Cycle 1 plane: [0.99, -0.07, -0.12], residual 0.254
Cycle 2 plane: [-0.64, 0.40, 0.65], residual 0.389

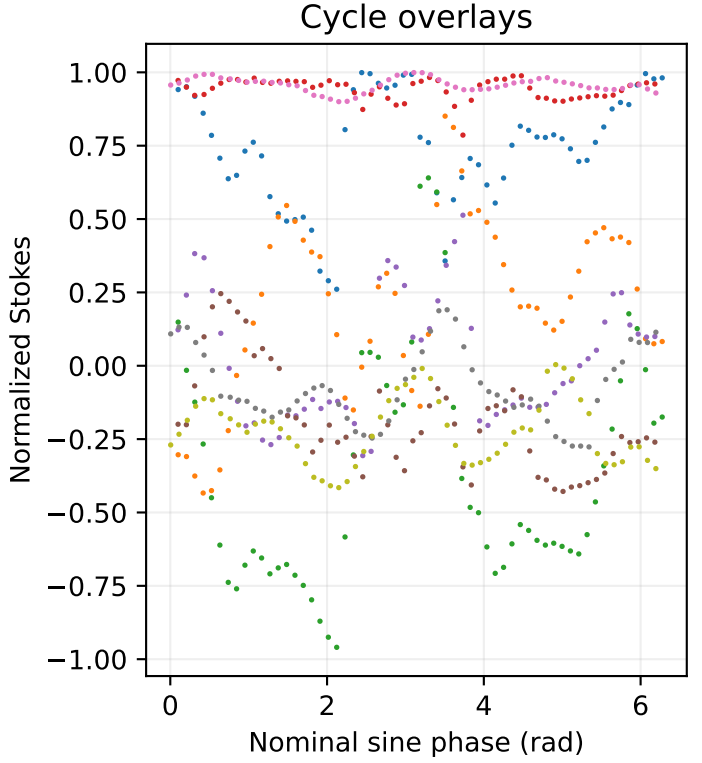
phi1 sine; fixed phi2 = 0.5425 RP V (segment 7)





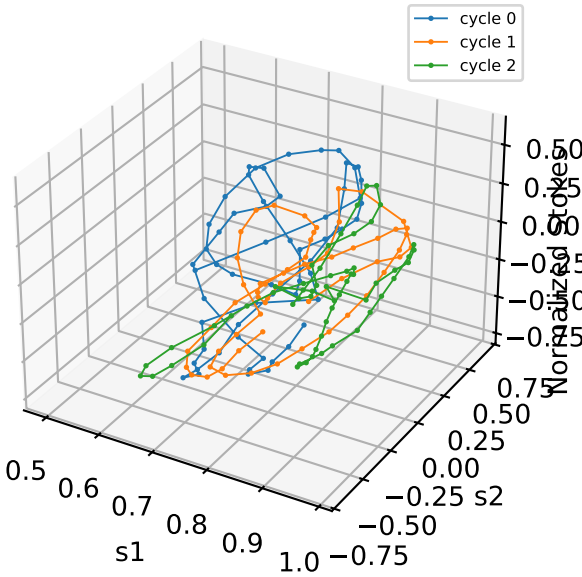


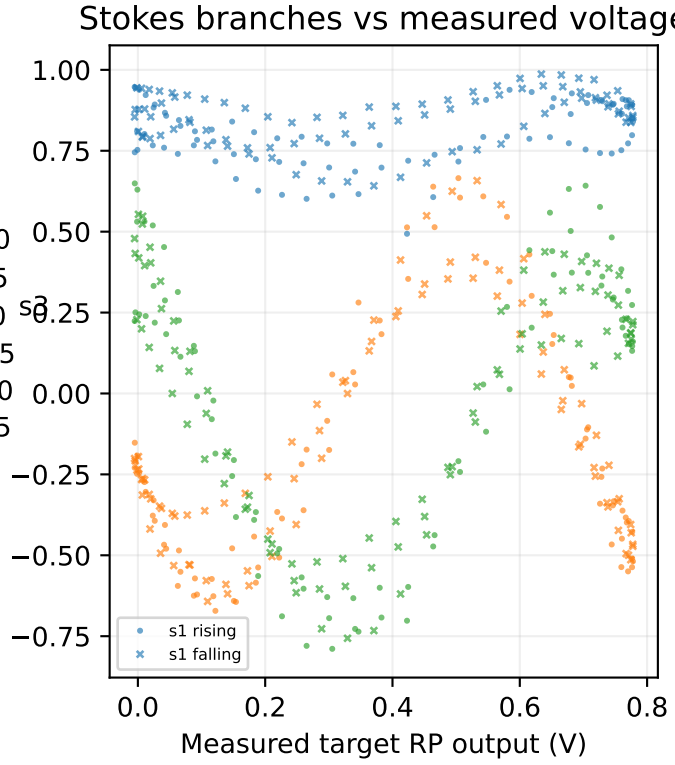


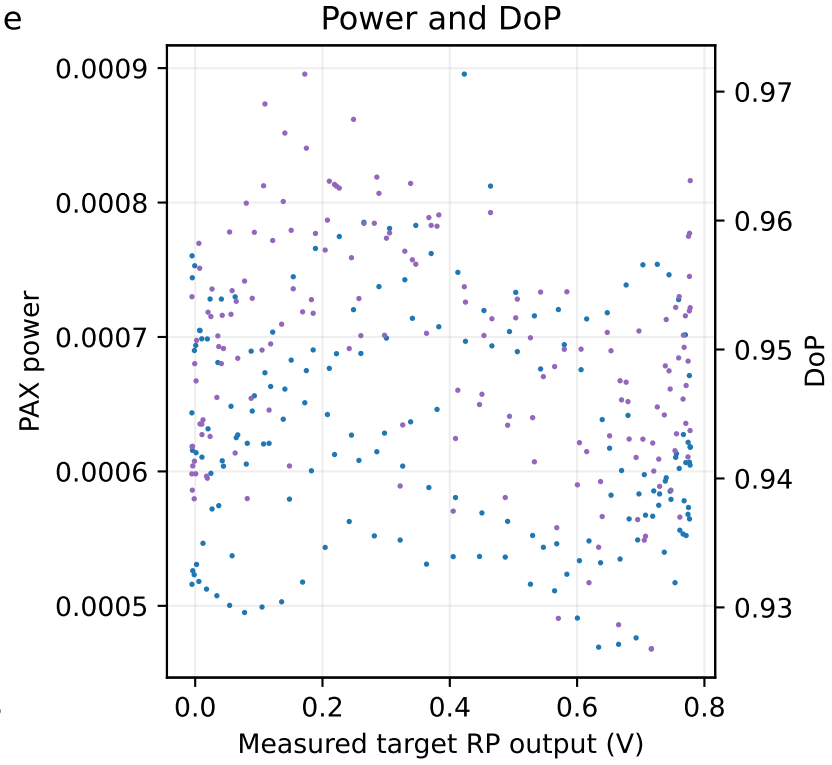


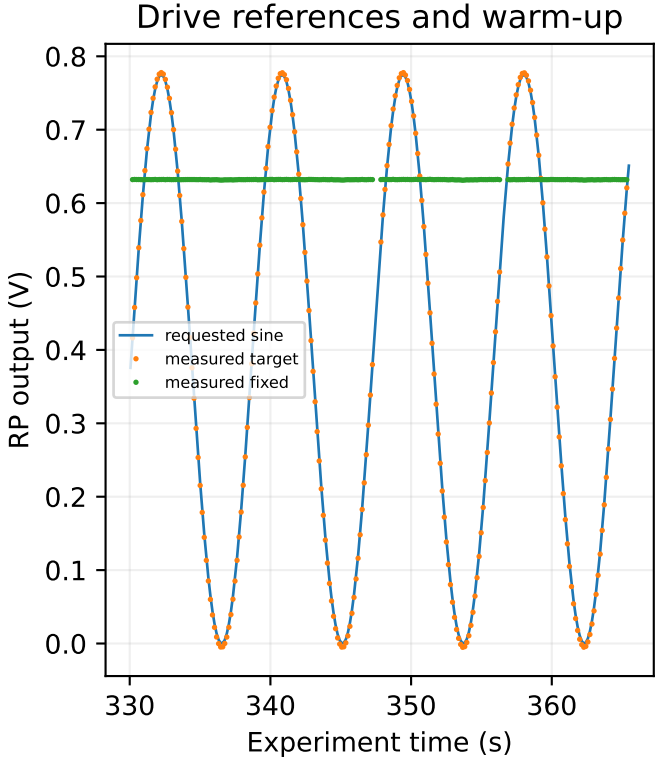
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.917, 0.295, -0.268]
Plane residual RMS: 0.09602
Rising/falling angular RMS: 23.4°
Rising/falling power RMS: 8.52e-05 PAX units
Cycle angular RMS: 39.4°
Cycle power RMS: 0.000114 PAX units
Cycle DoP RMS: 0.0658
Tracking RMS: 0.003041 V
Reference coverage: 96.0%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.92, 0.31, -0.24], residual 0.131
Cycle 1 plane: [0.99, 0.07, -0.11], residual 0.026
Cycle 2 plane: [0.98, -0.05, -0.19], residual 0.0102

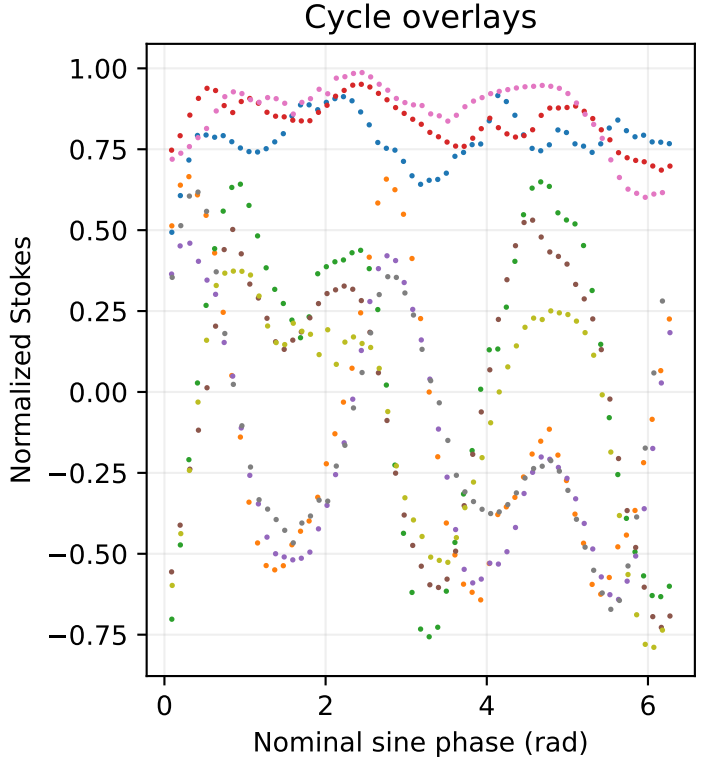
phi1 sine; fixed phi2 = 0.62 RP V (segment 8)





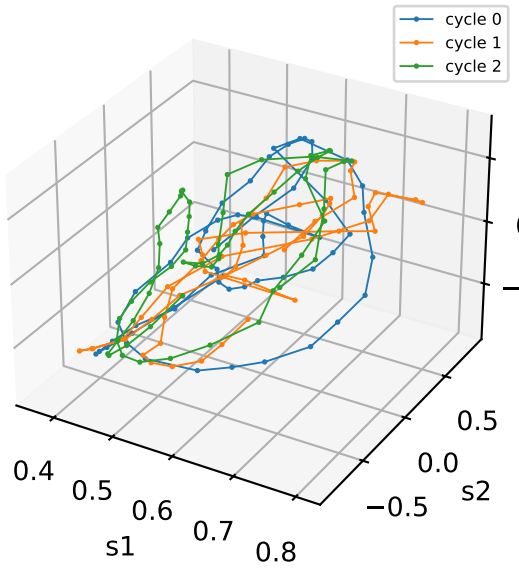


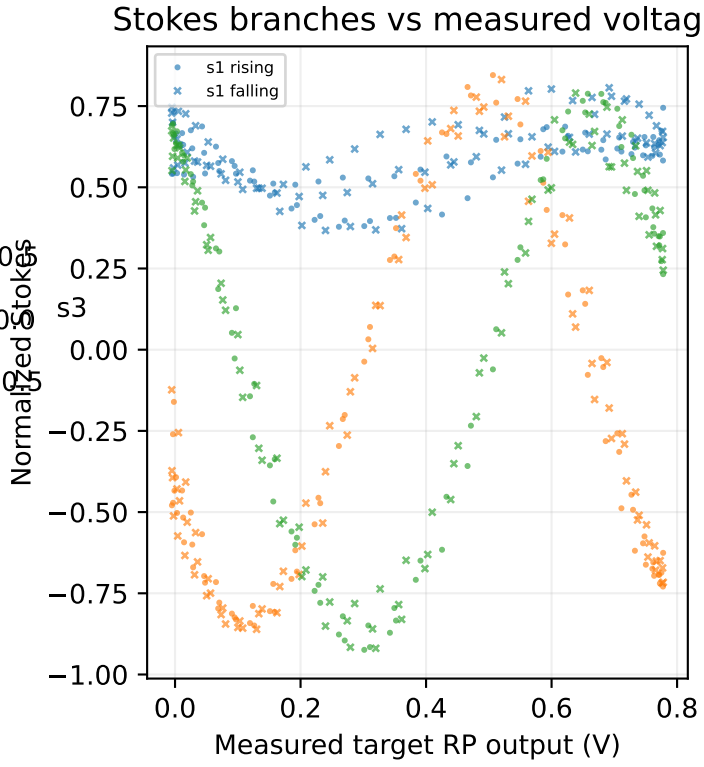


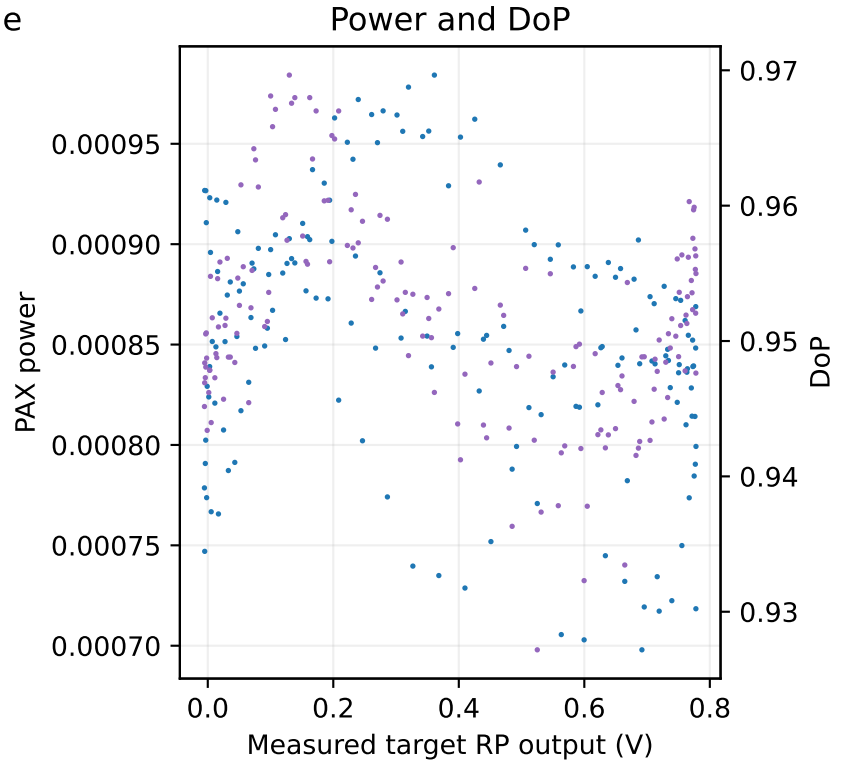


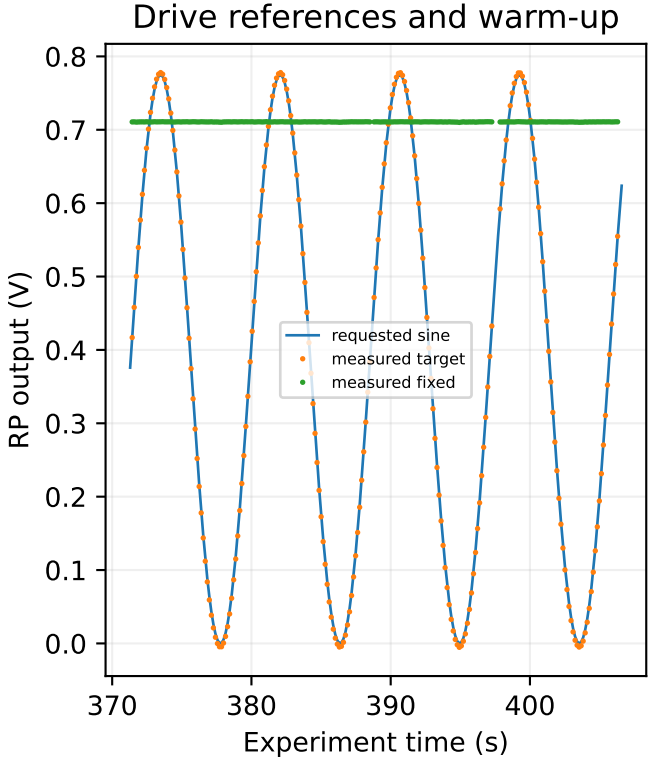
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.992, -0.016, -0.125]
Plane residual RMS: 0.07773
Rising/falling angular RMS: 12.3°
Rising/falling power RMS: 0.000119 PAX units
Cycle angular RMS: 14.4°
Cycle power RMS: 0.000132 PAX units
Cycle DoP RMS: 0.00744
Tracking RMS: 0.00309 V
Reference coverage: 96.6%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [1.00, 0.04, -0.09], residual 0.0645
Cycle 1 plane: [0.98, -0.10, -0.14], residual 0.0381
Cycle 2 plane: [0.97, -0.03, -0.23], residual 0.0671

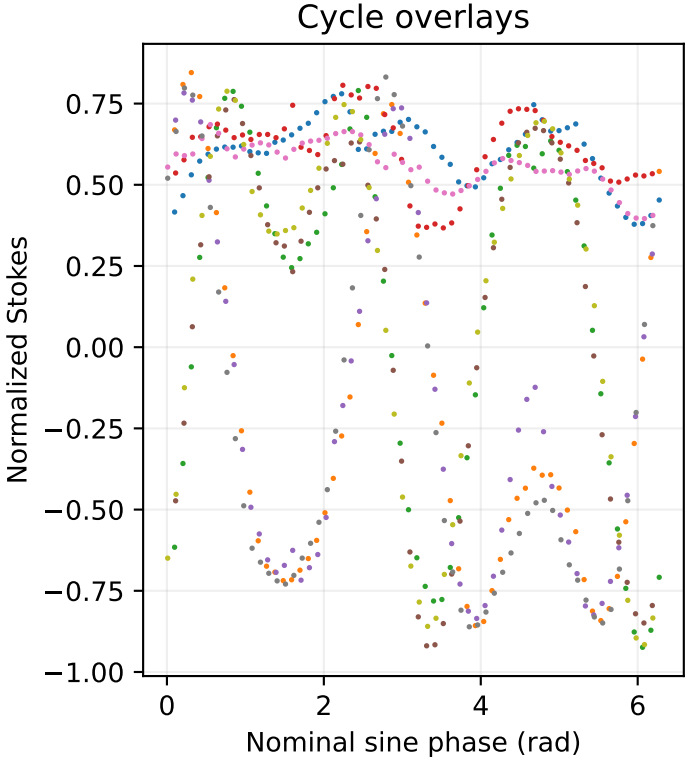
phi1 sine; fixed phi2 = 0.6974 RP V (segment 9)





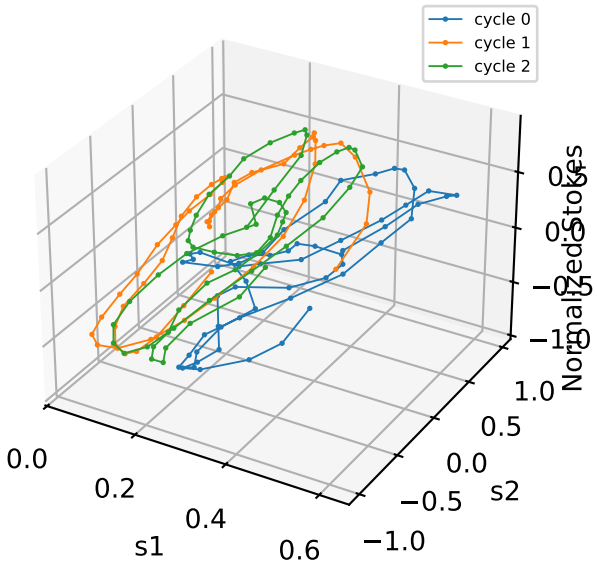


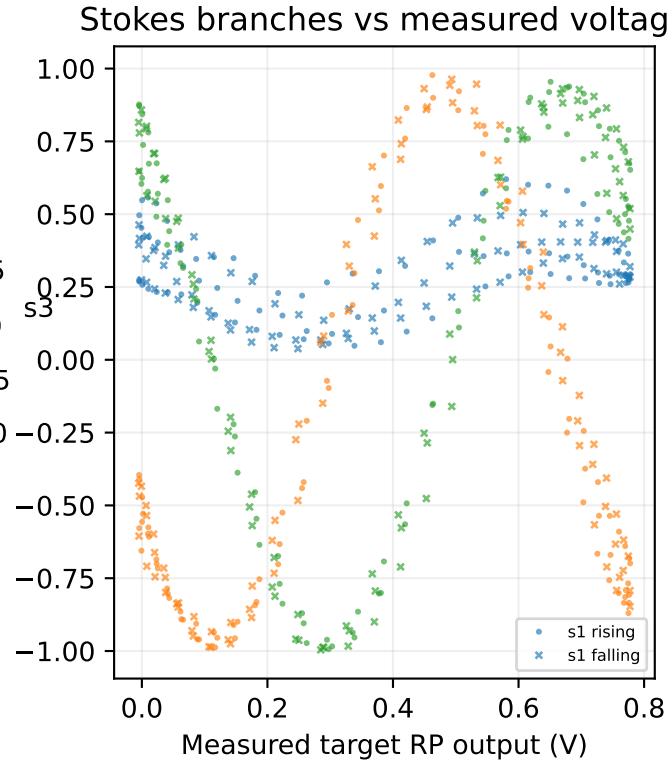


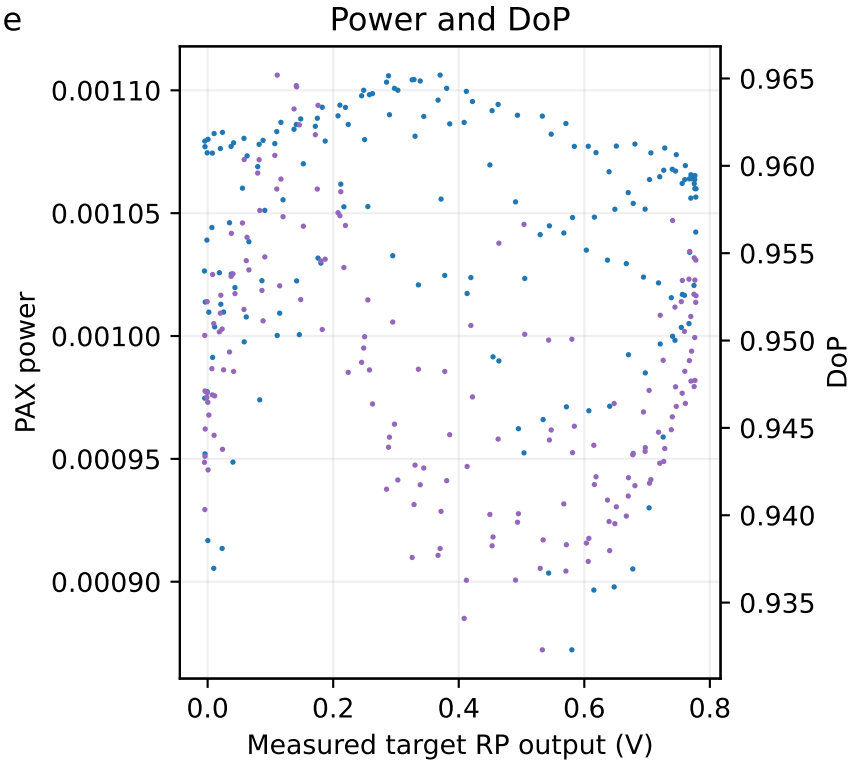


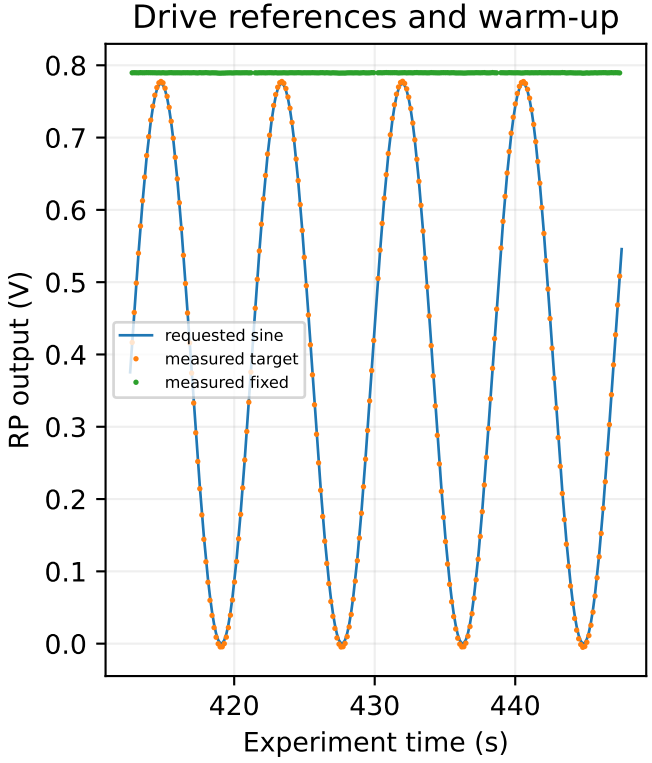
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.990, -0.038, -0.135]
Plane residual RMS: 0.06585
Rising/falling angular RMS: 10.6°
Rising/falling power RMS: 9.57e-05 PAX units
Cycle angular RMS: 9.49°
Cycle power RMS: 8.34e-05 PAX units
Cycle DoP RMS: 0.00404
Tracking RMS: 0.003003 V
Reference coverage: 97.7%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, -0.01, -0.12], residual 0.077
Cycle 1 plane: [0.98, -0.05, -0.18], residual 0.0538
Cycle 2 plane: [0.99, -0.05, -0.10], residual 0.0386

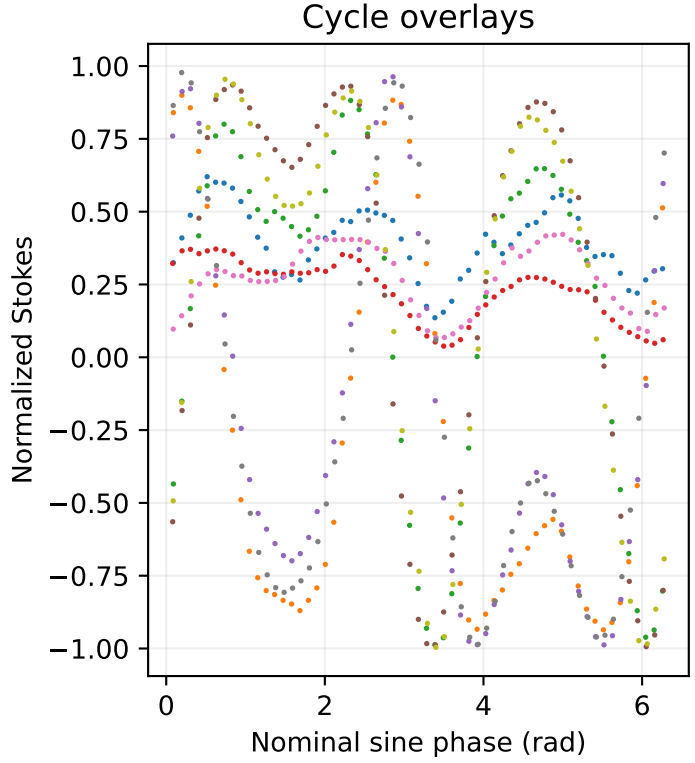
phi1 sine; fixed phi2 = 0.7749 RP V (segment 10)









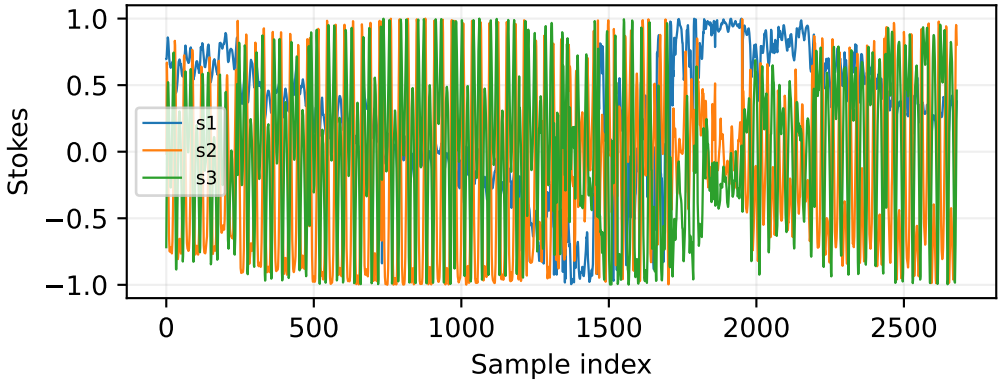


Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.989, -0.039, -0.146]
Plane residual RMS: 0.09417
Rising/falling angular RMS: 8.01°
Rising/falling power RMS: 3.79e-05 PAX units
Cycle angular RMS: 14.6°
Cycle power RMS: 8.63e-05 PAX units
Cycle DoP RMS: 0.00443
Tracking RMS: 0.003065 V
Reference coverage: 98.3%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.98, -0.07, -0.17], residual 0.0585
Cycle 1 plane: [0.99, -0.07, -0.14], residual 0.0371
Cycle 2 plane: [0.99, -0.00, -0.15], residual 0.0546

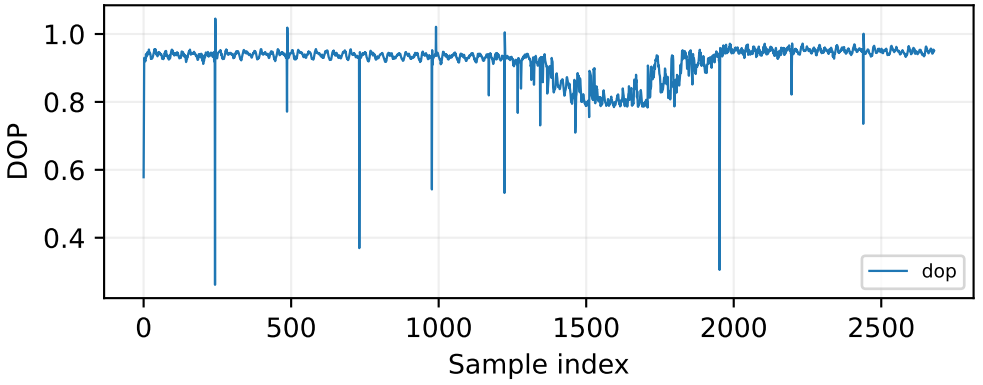
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

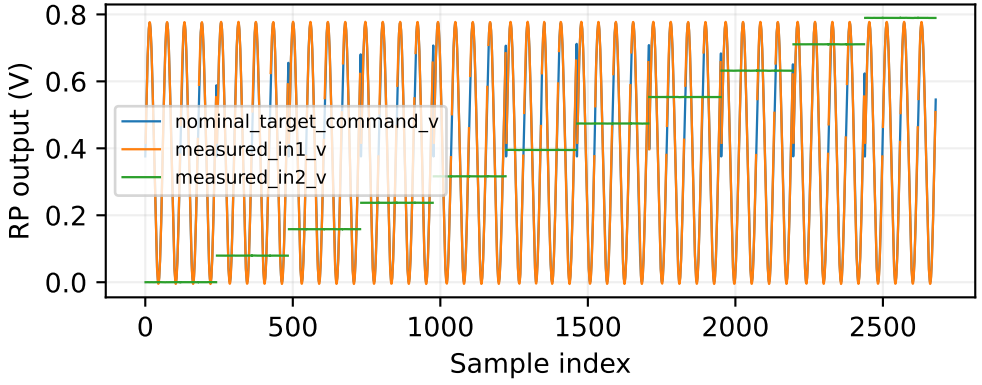
Normalized Stokes



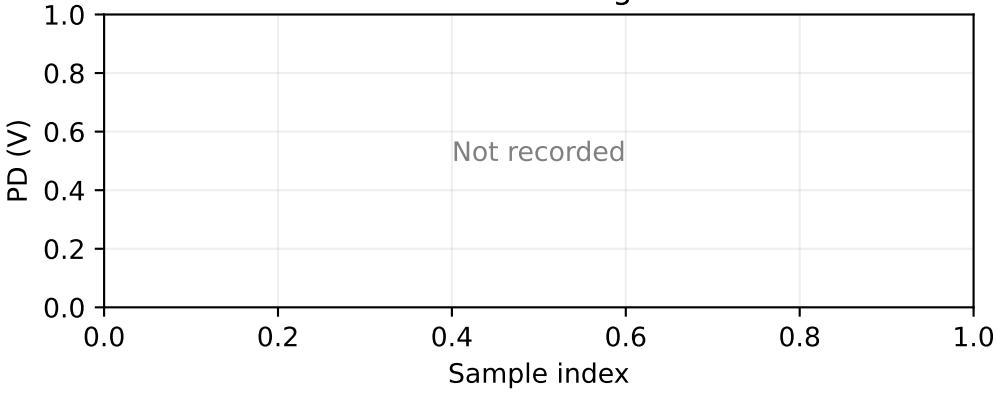
Raw degree of polarization



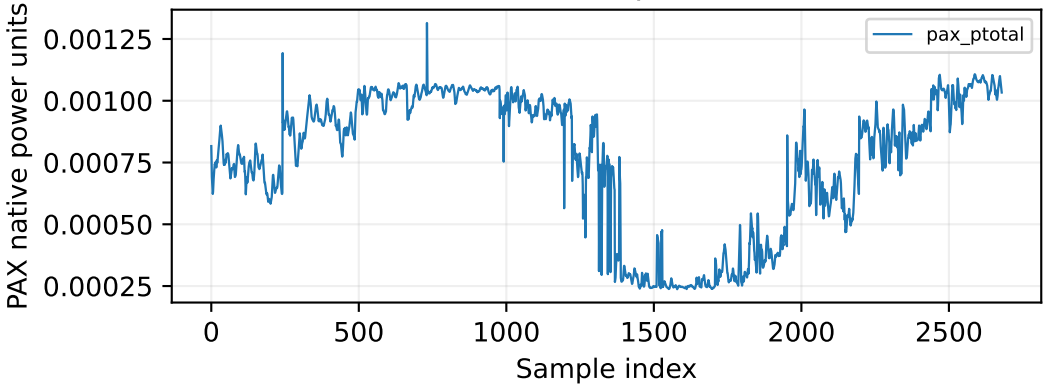
Actuator command / reference history



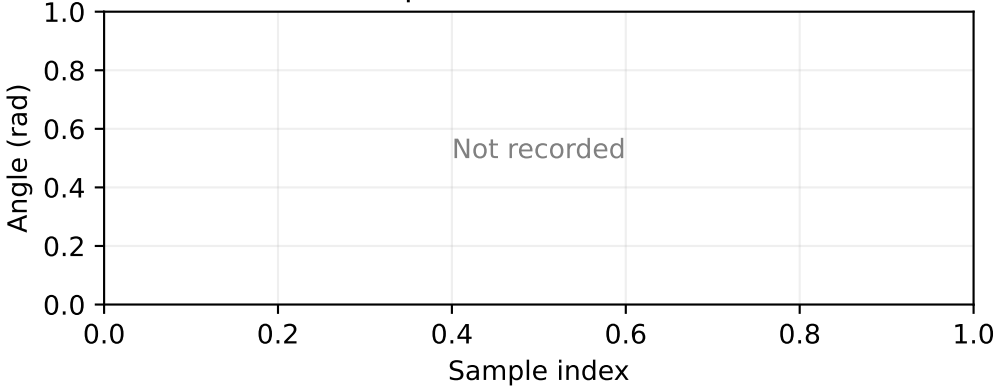
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"phi1-rerun"
comment	"another re run for the sake of having data to analyze later. time
actuator	"phi1"
report	"pdf"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	13.078125
phi2_v_lambda	13.078125
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	16.875
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0
cross_sweep_sine_frequency_hz	0.1
cross_sweep_sine_center_voltage	0.3875
cross_sweep_sine_amplitude_voltage	0.3875
cross_sweep_warmup_cycles	1
cross_sweep_recorded_cycles	3
cross_sweep_phi1_bias_voltages	[0.0, 0.0775, 0.155, 0.2325, 0.31, 0.3875, 0.465, 0.5425, 0.62, 0
cross_sweep_phi2_bias_voltages	[0.0, 0.0775, 0.155, 0.2325, 0.31, 0.3875, 0.465, 0.5425, 0.62, 0
cross_sweep_settle_s	1.0
cross_sweep_bias_ramp_s	1.0
cross_sweep_bias_ramp_updates_per_s	50.0
cross_sweep_reference_sample_rate_hz	400.0

Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
cross_sweep_scope_block_s	8.0
cross_sweep_min_pax_samples_per_cycle	20.0
cross_sweep_pax_latency_s	0.12